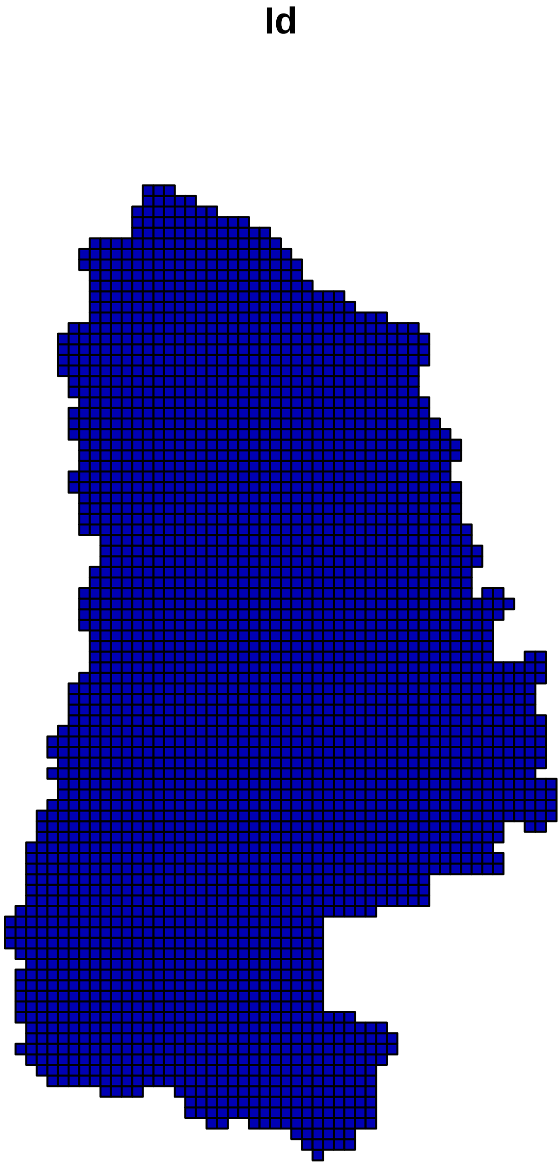
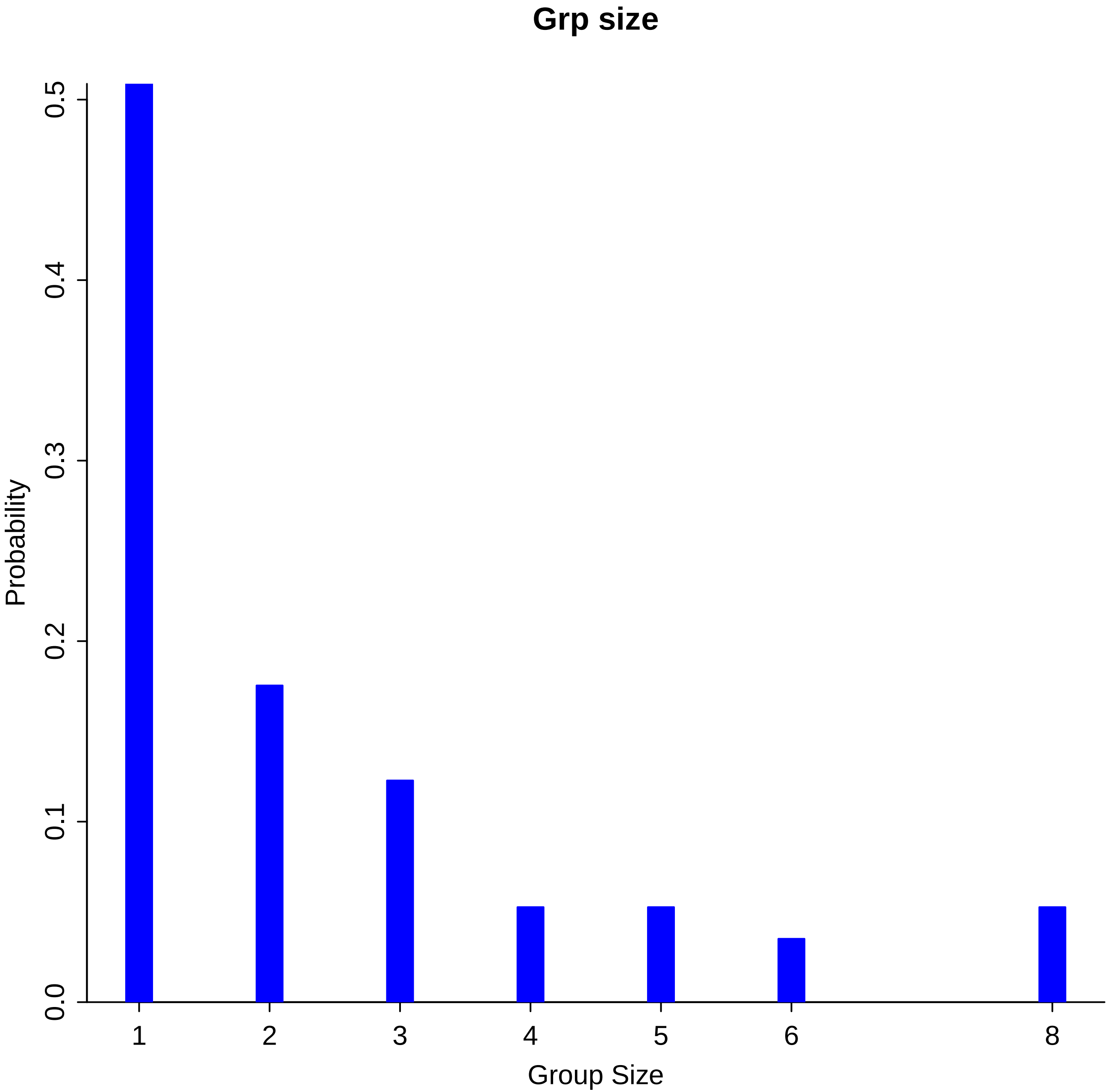
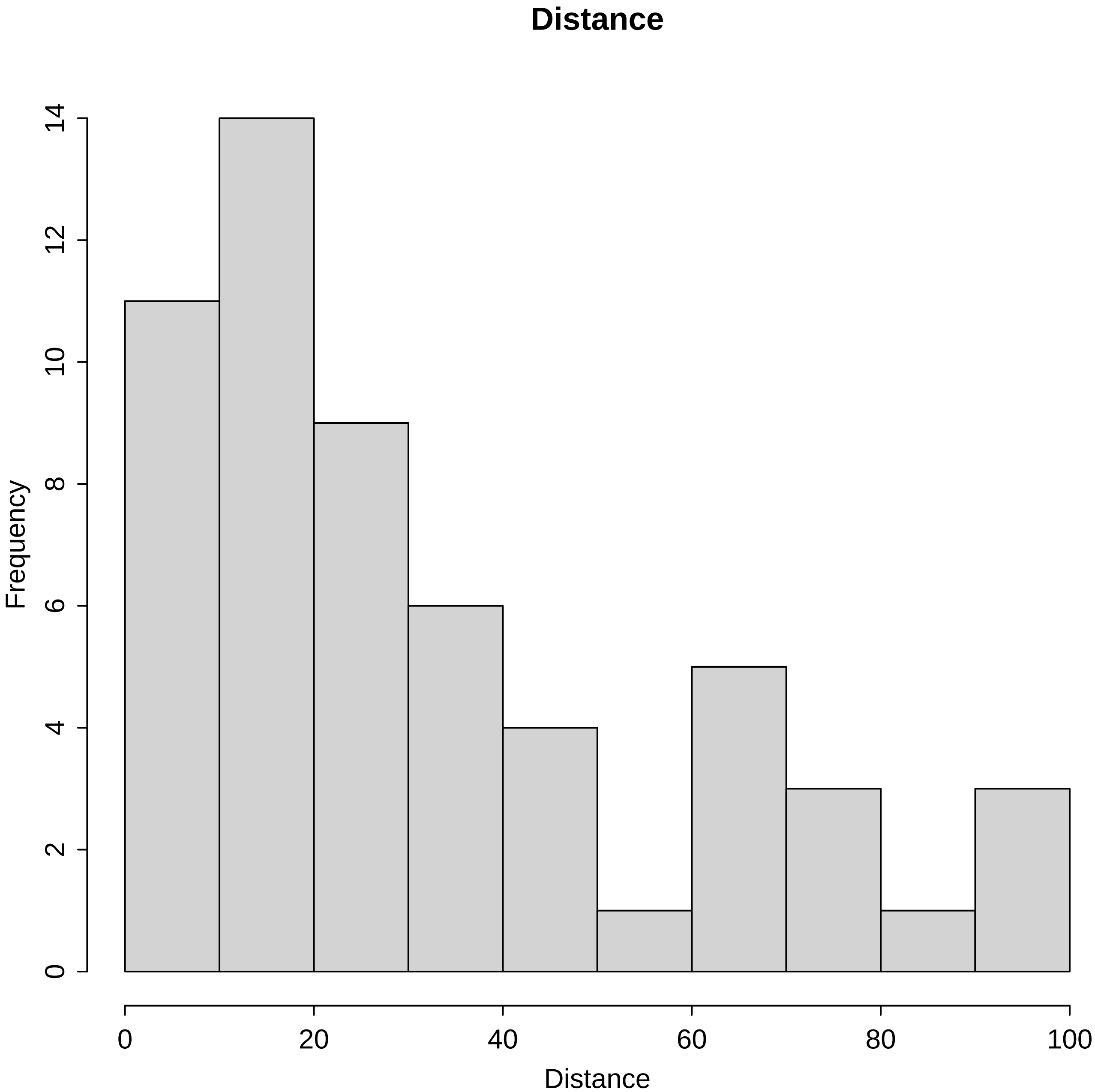
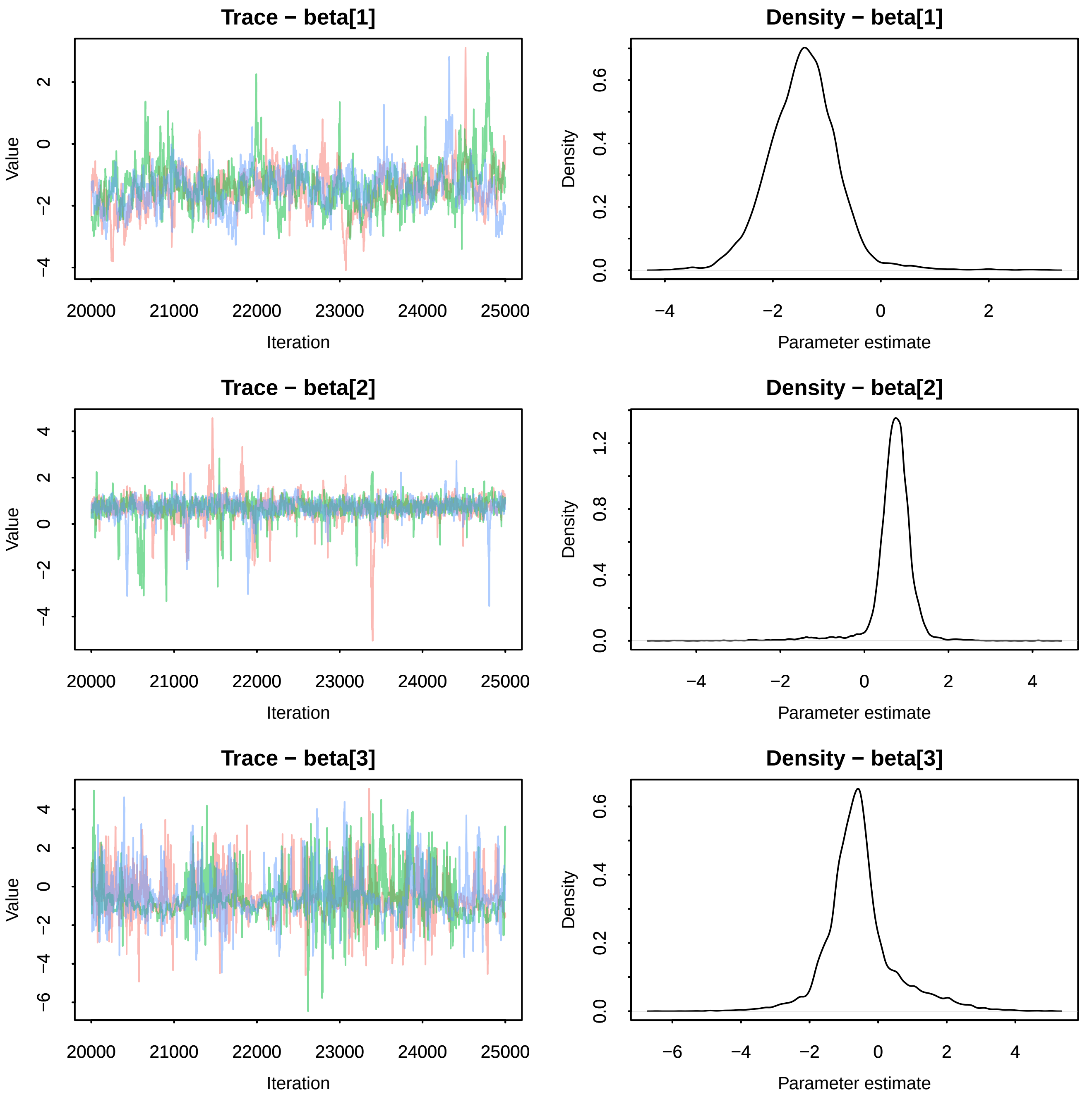
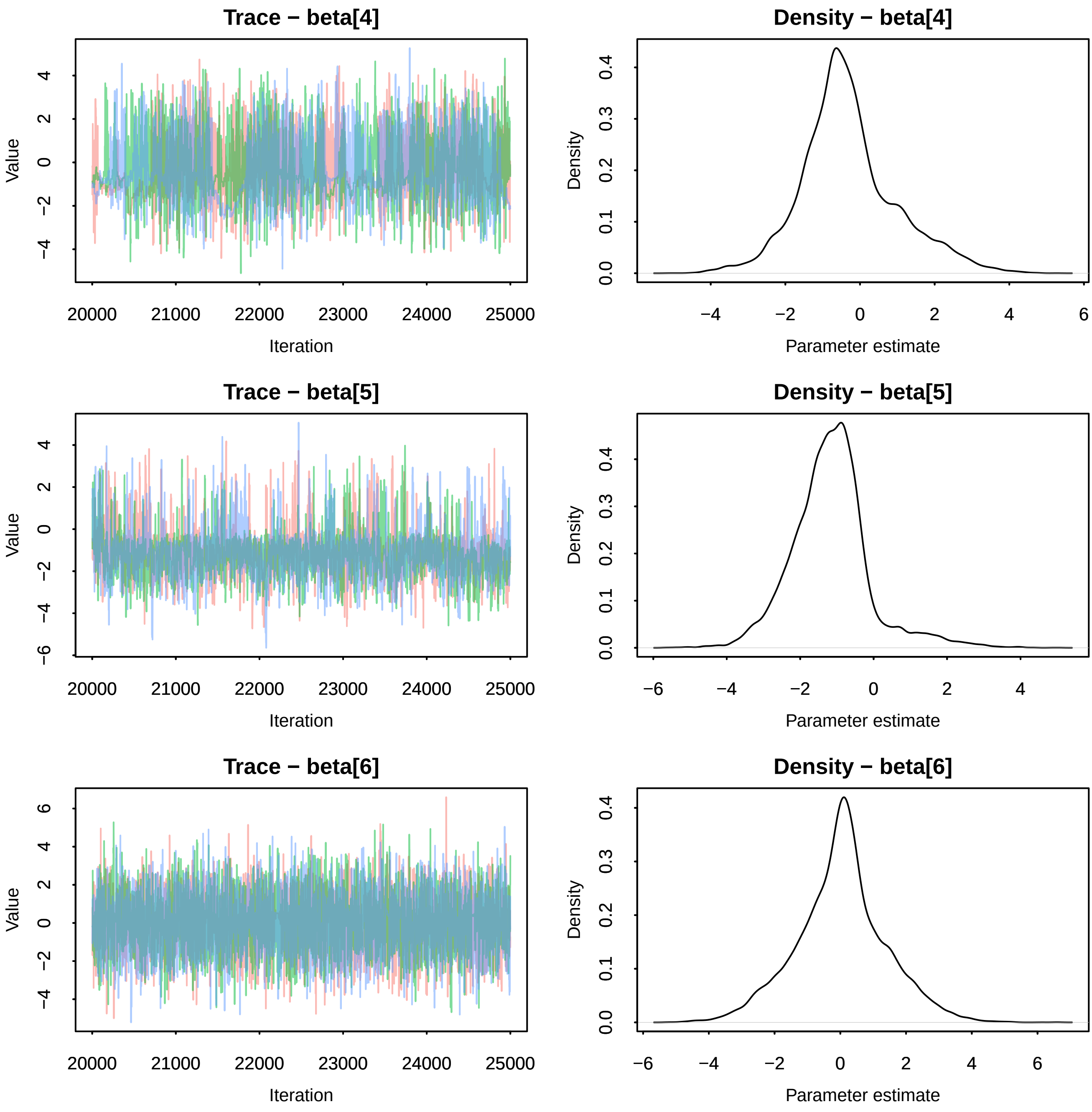
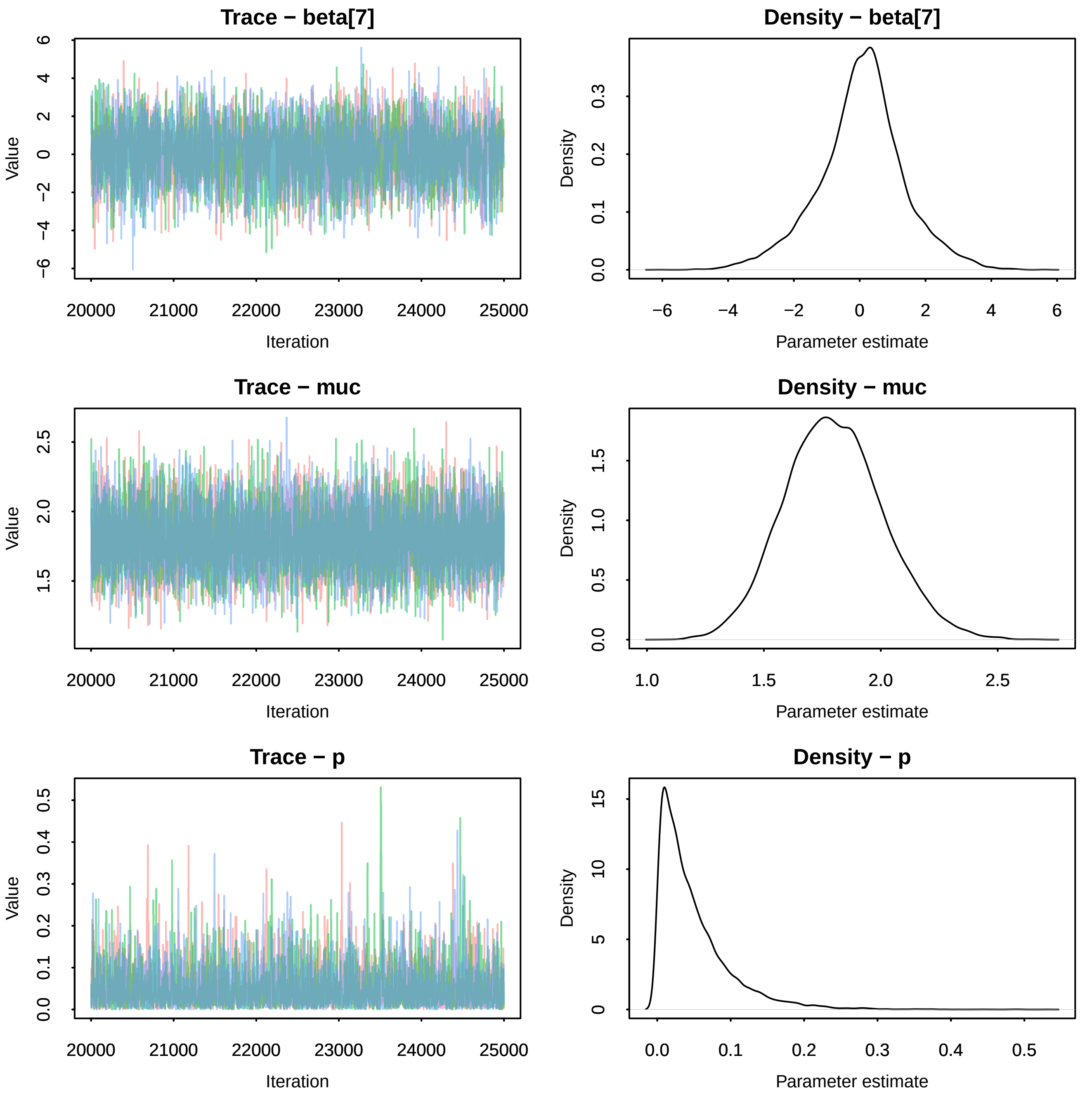

Id
ID

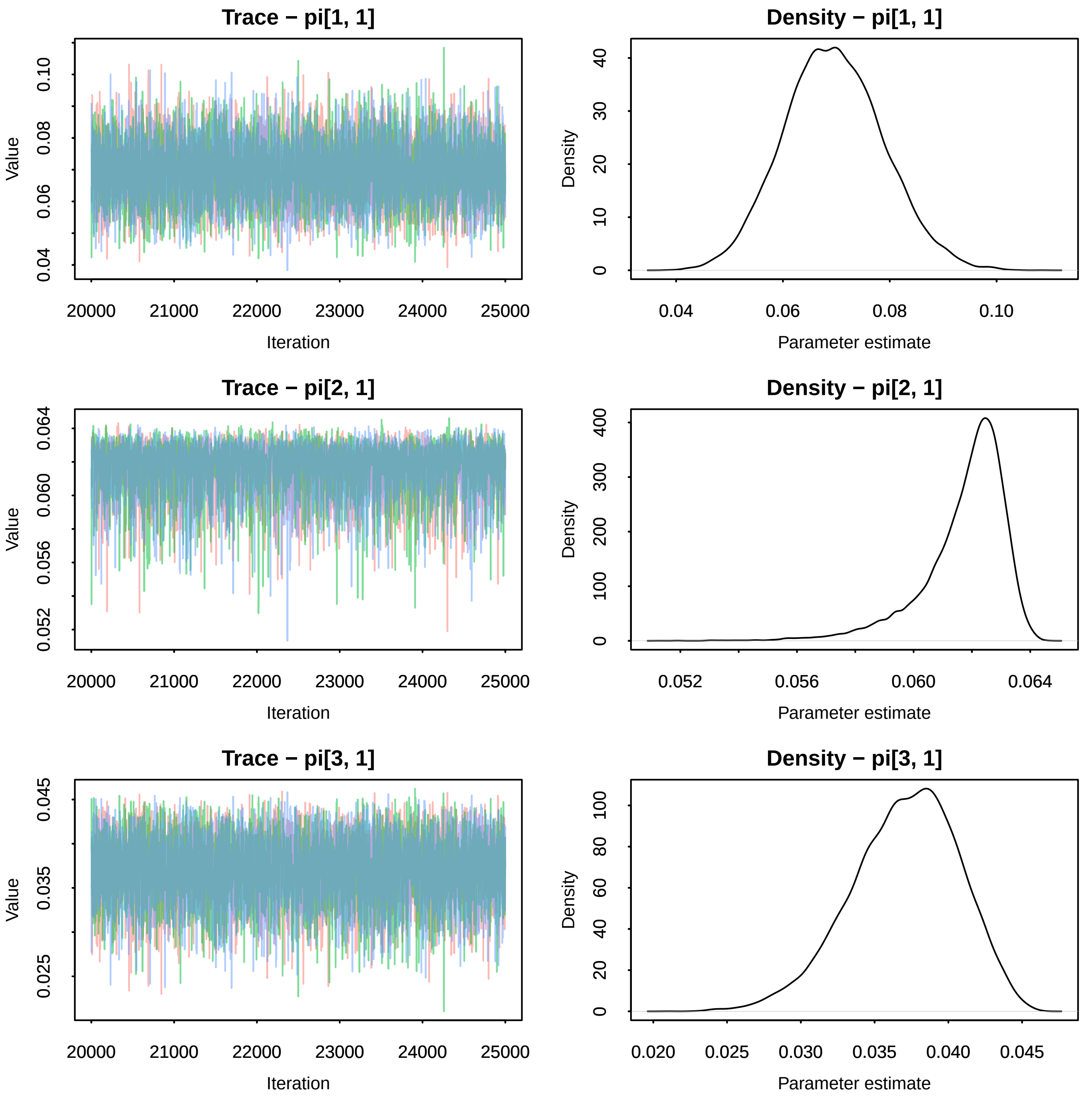

Grp size
Probability
0.0
0.1
0.2
0.3
0.4
0.5
1
2
3
4
5
6
8
Group Size

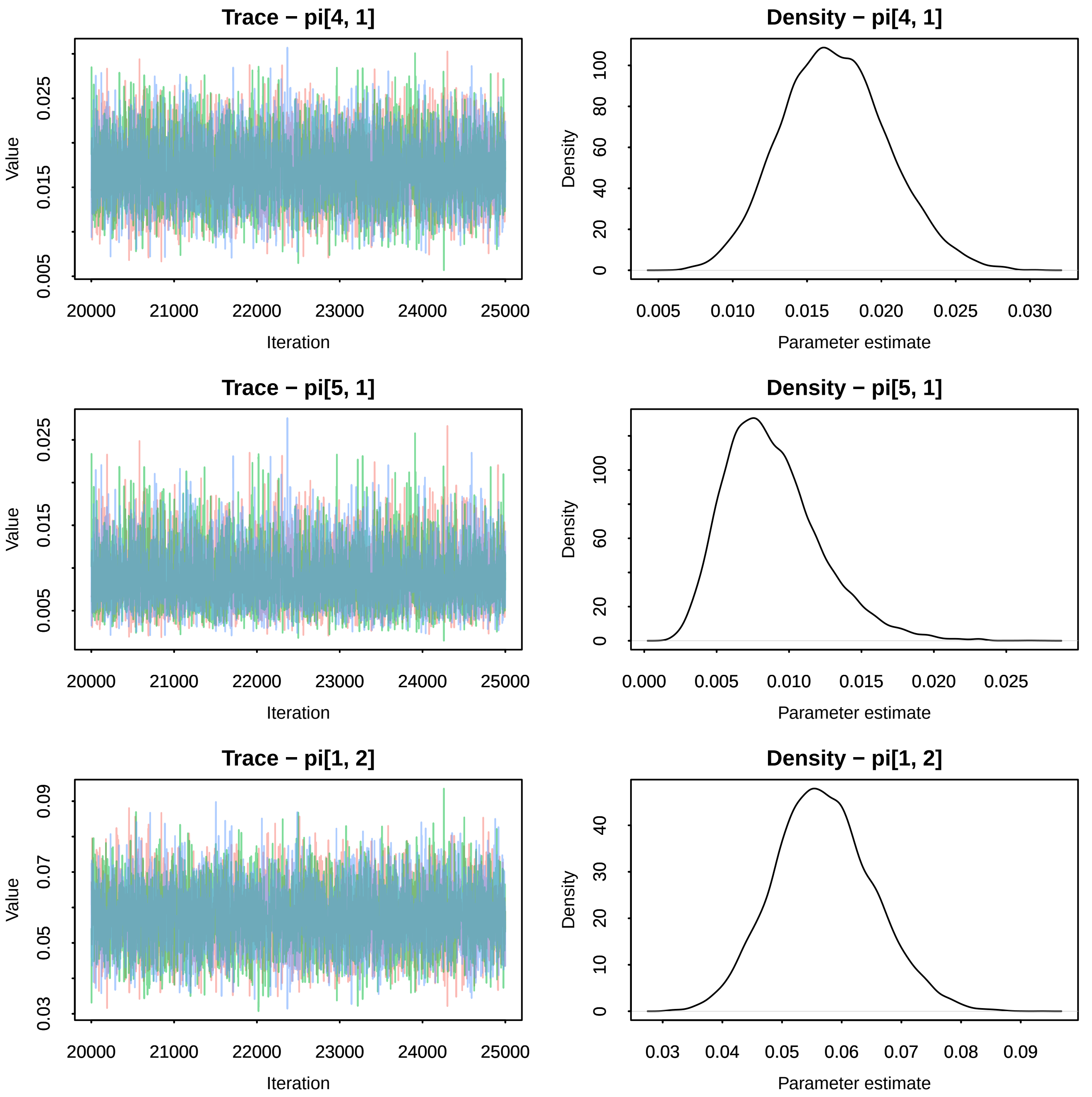

Distance
Frequency
0
2
4
6
8
10
12
14
0
20
40
60
80
100
Distance


Trace − beta[1]
Density − beta[1]
Trace − beta[2]
Density − beta[2]
Trace − beta[3]
Density − beta[3]
Value
Iteration
Density
Parameter estimate

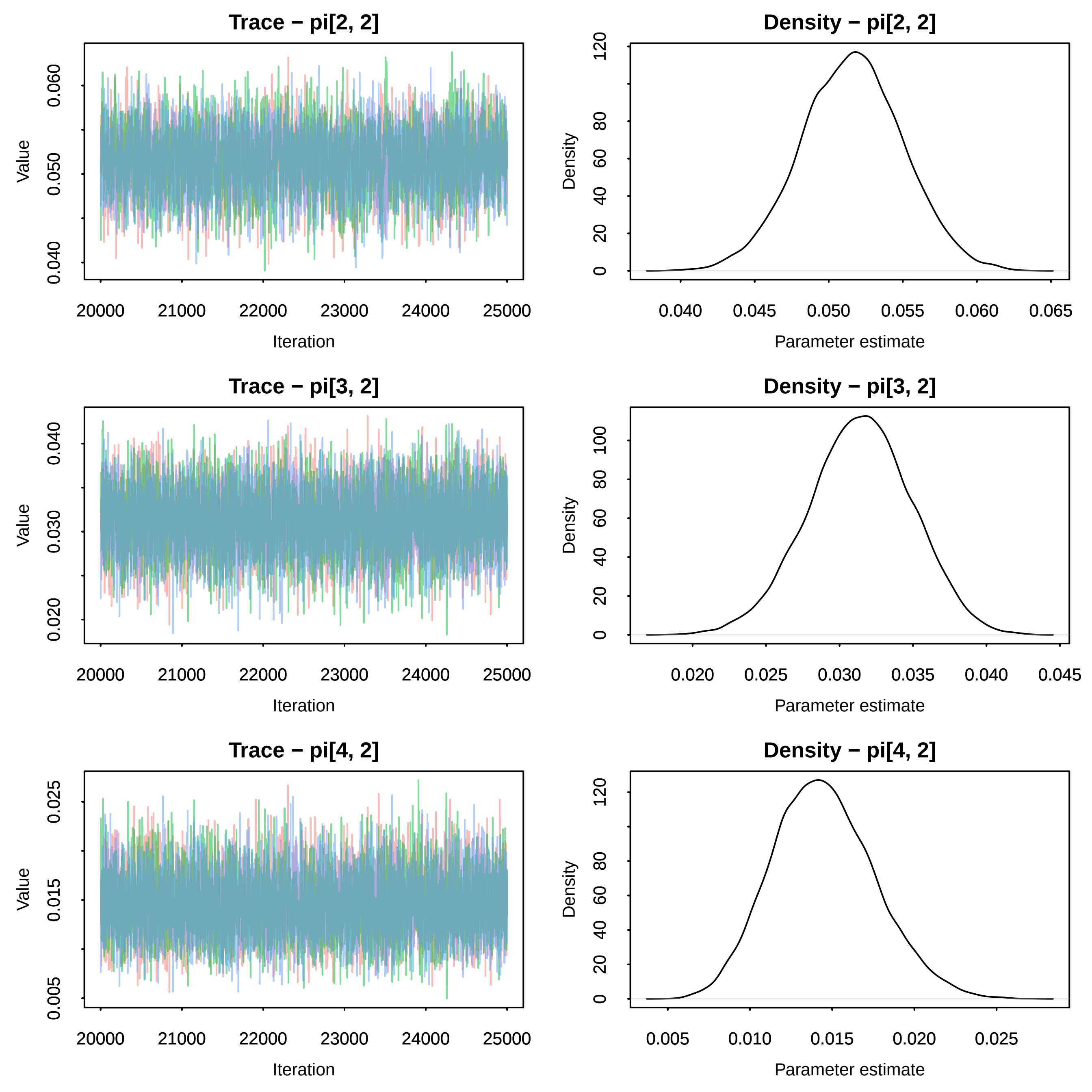

Trace − beta[4]
Density − beta[4]
Trace − beta[5]
Density − beta[5]
Trace − beta[6]
Density − beta[6]
Value
Iteration
Density
Parameter estimate

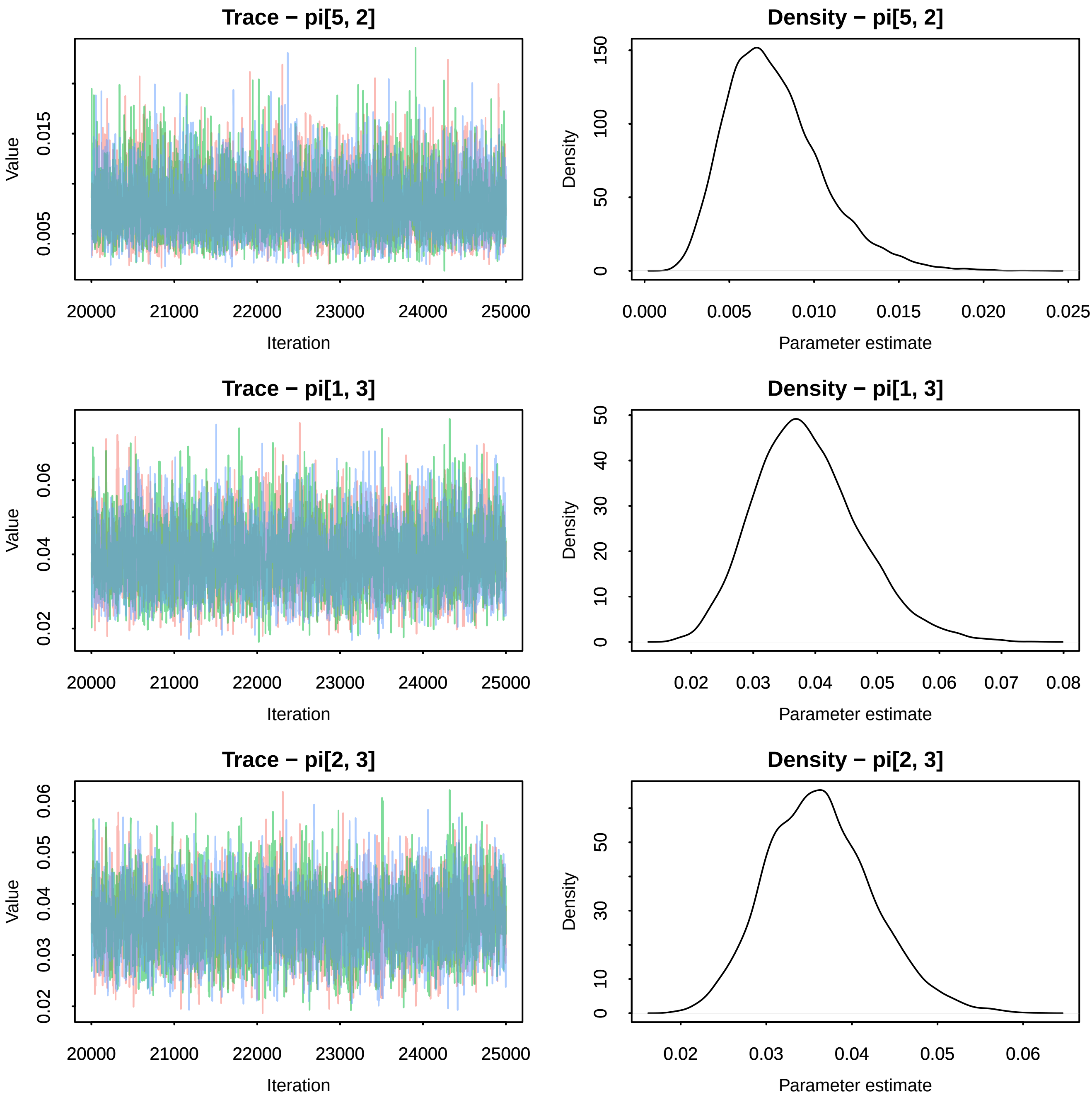

Trace − beta[7]
Density − beta[7]
Trace − muc
Density − muc
Trace − p
Density − p
Value
Density
Iteration
Parameter estimate
20000
21000
22000
23000
24000
25000

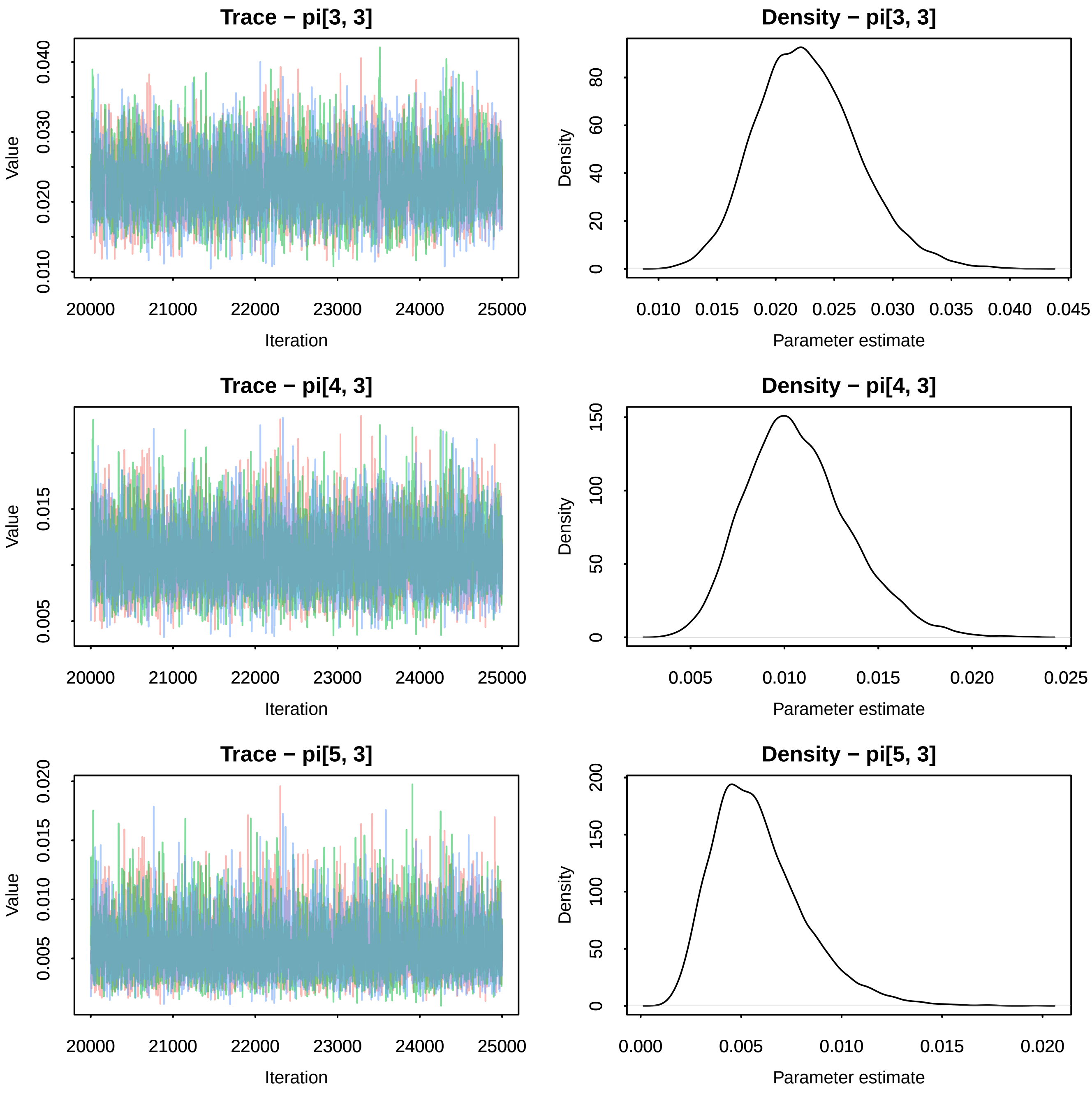

Trace − pi[1, 1]
Density − pi[1, 1]
Value
Iteration
Density
Parameter estimate
20000
21000
22000
23000
24000
25000
0.04
0.06
0.08
0.10
0
10
20
30
40
Trace − pi[2, 1]
Density − pi[2, 1]
Value
Iteration
Density
Parameter estimate
0.052
0.056
0.060
0.064
0
100
200
300
400
Trace − pi[3, 1]
Density − pi[3, 1]
Value
Iteration
Density
Parameter estimate
0.025
0.035
0.045
0.020
0.030
0.040
0
20
40
60
80
100



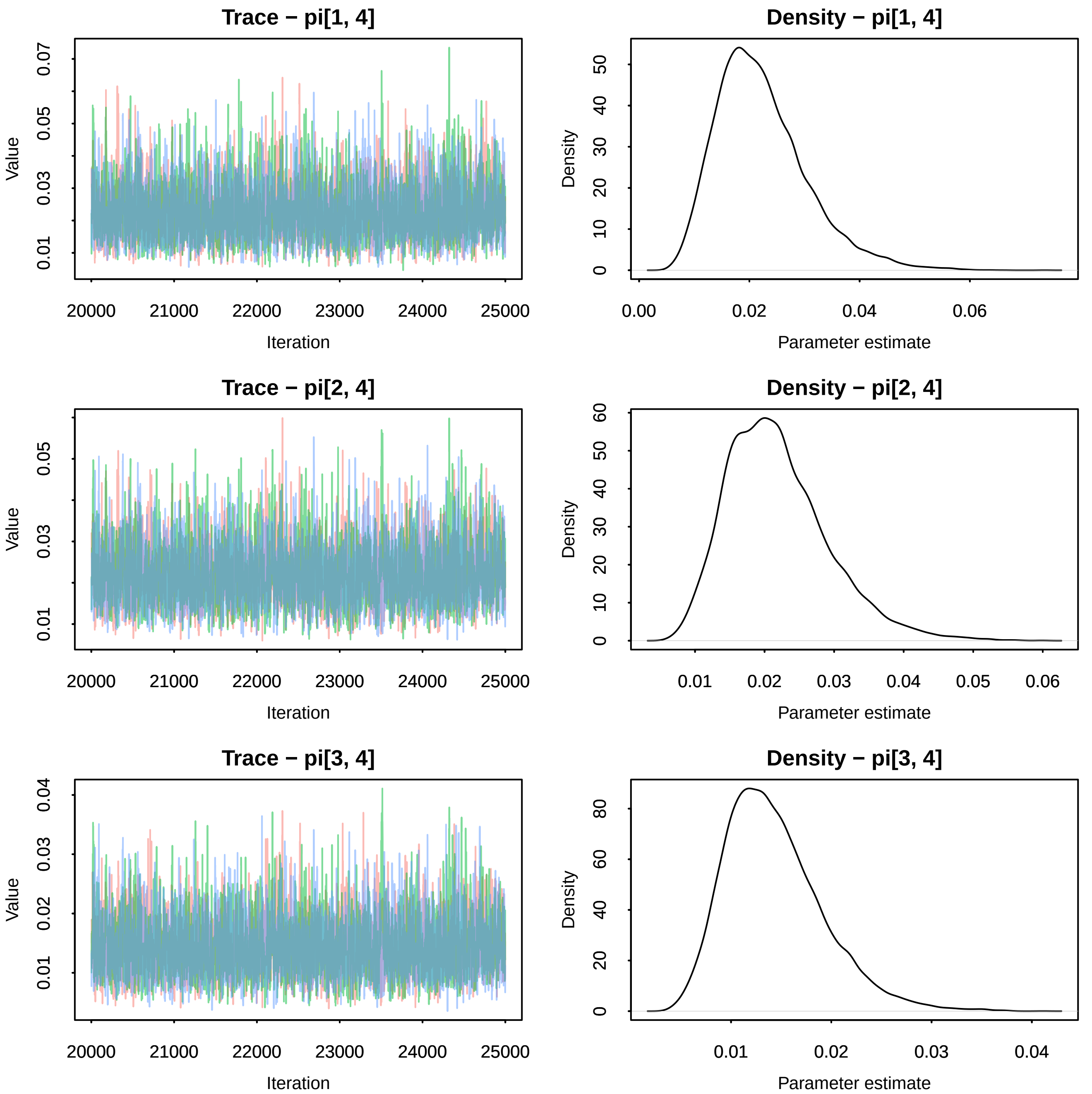
Trace − pi[4, 1]
Value
0.005
0.015
0.025
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[4, 1]
Density
0
20
40
60
80
100
0.005
0.010
0.015
0.020
0.025
0.030
Parameter estimate
Trace − pi[5, 1]
Value
0.005
0.015
0.025
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[5, 1]
Density
0
20
60
100
0.000
0.005
0.010
0.015
0.020
0.025
Parameter estimate
Trace − pi[1, 2]
Value
0.03
0.05
0.07
0.09
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[1, 2]
Density
0
10
20
30
40
0.03
0.04
0.05
0.06
0.07
0.08
0.09
Parameter estimate


Trace − pi[2, 2]
Value
0.040
0.050
0.060
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[2, 2]
Density
0
20
40
60
80
120
0.040
0.045
0.050
0.055
0.060
0.065
Parameter estimate
Trace − pi[3, 2]
Value
0.020
0.030
0.040
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[3, 2]
Density
0
20
40
60
80
100
0.020
0.025
0.030
0.035
0.040
0.045
Parameter estimate
Trace − pi[4, 2]
Value
0.005
0.015
0.025
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[4, 2]
Density
0
20
40
60
80
120
0.005
0.010
0.015
0.020
0.025
Parameter estimate

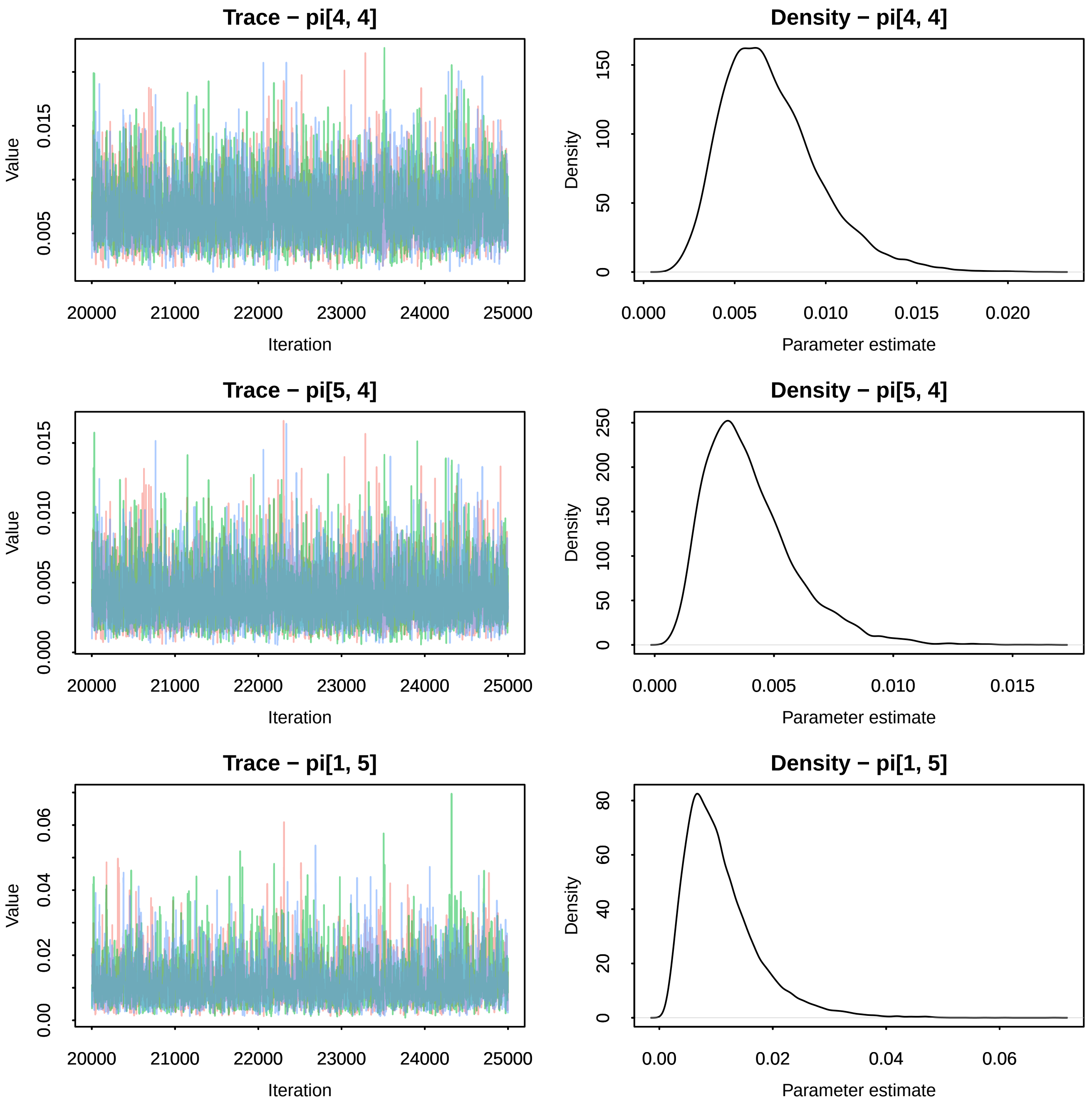

Trace − pi[5, 2]
Value
0.005
0.015
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[5, 2]
Density
0
50
100
150
0.000
0.005
0.010
0.015
0.020
0.025
Parameter estimate
Trace − pi[1, 3]
Value
0.02
0.04
0.06
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[1, 3]
Density
0
10
20
30
40
50
0.02
0.03
0.04
0.05
0.06
0.07
0.08
Parameter estimate
Trace − pi[2, 3]
Value
0.02
0.03
0.04
0.05
0.06
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[2, 3]
Density
0
10
30
50
0.02
0.03
0.04
0.05
0.06
Parameter estimate

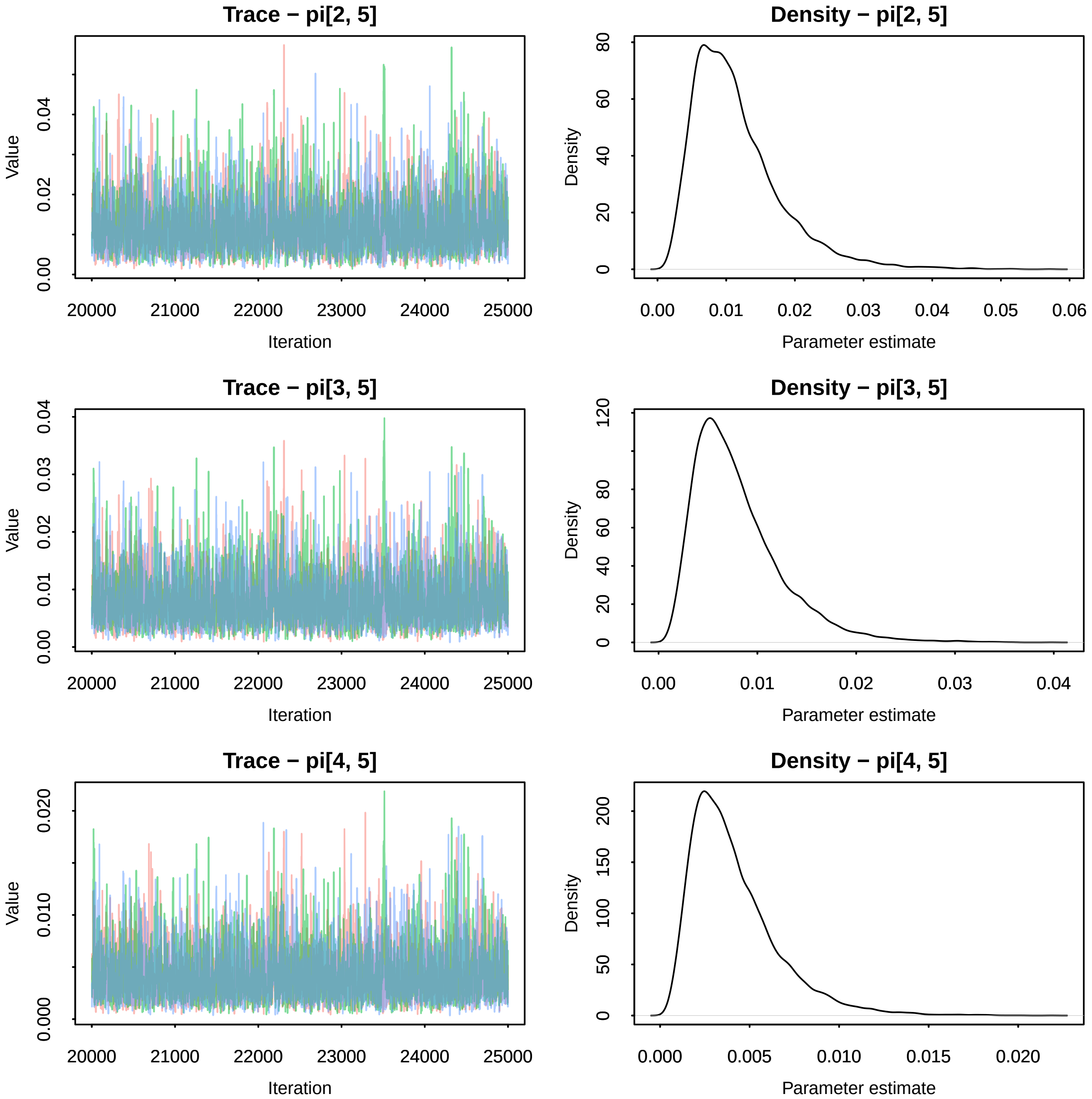


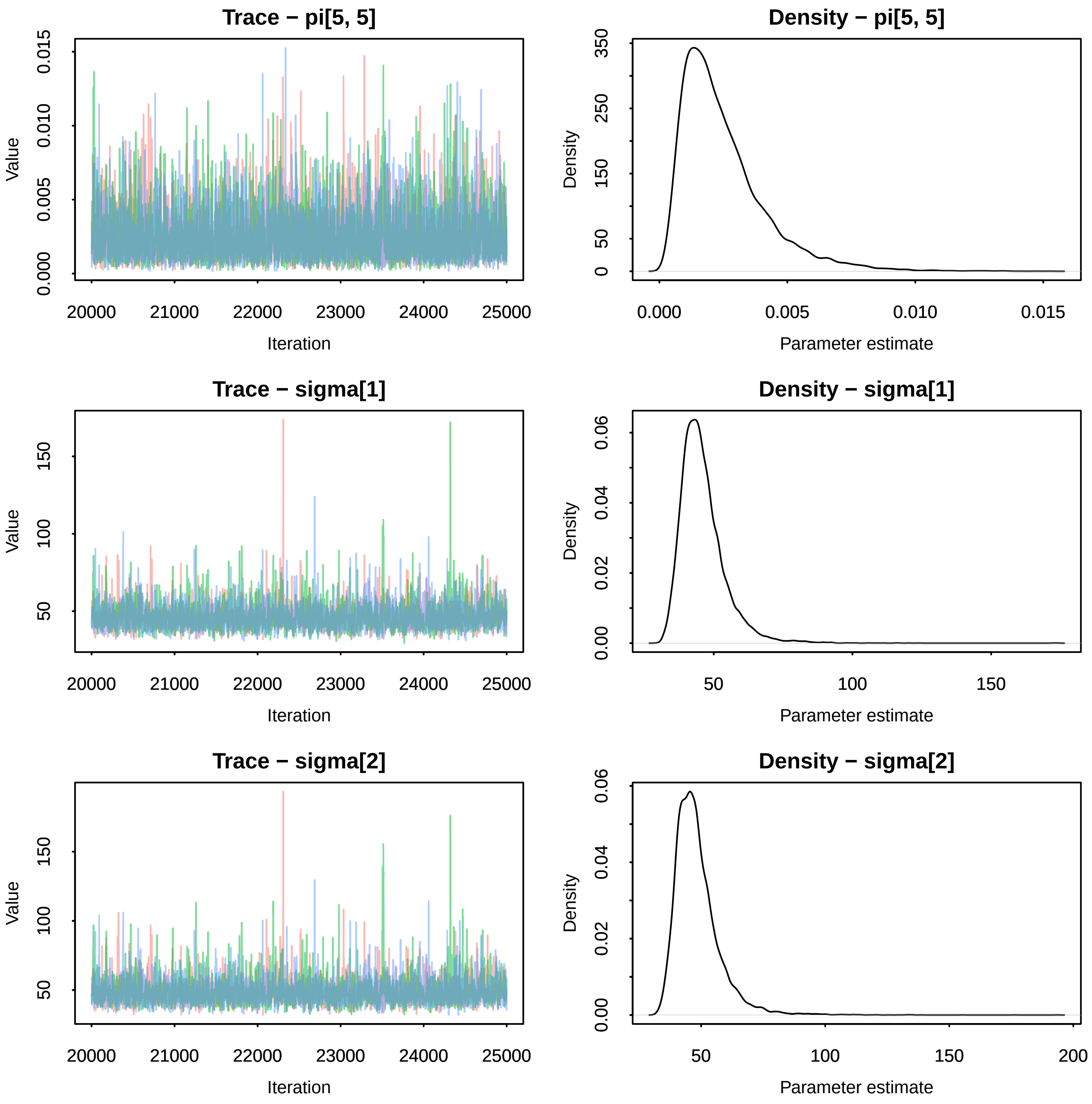
Trace − pi[3, 3]
Value
0.010
0.020
0.030
0.040
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[3, 3]
Density
0
20
40
60
80
0.010
0.015
0.020
0.025
0.030
0.035
0.040
0.045
Parameter estimate
Trace − pi[4, 3]
Value
0.005
0.015
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[4, 3]
Density
0
50
100
150
0.005
0.010
0.015
0.020
0.025
Parameter estimate
Trace − pi[5, 3]
Value
0.005
0.010
0.015
0.020
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[5, 3]
Density
0
50
100
150
200
0.000
0.005
0.010
0.015
0.020
Parameter estimate

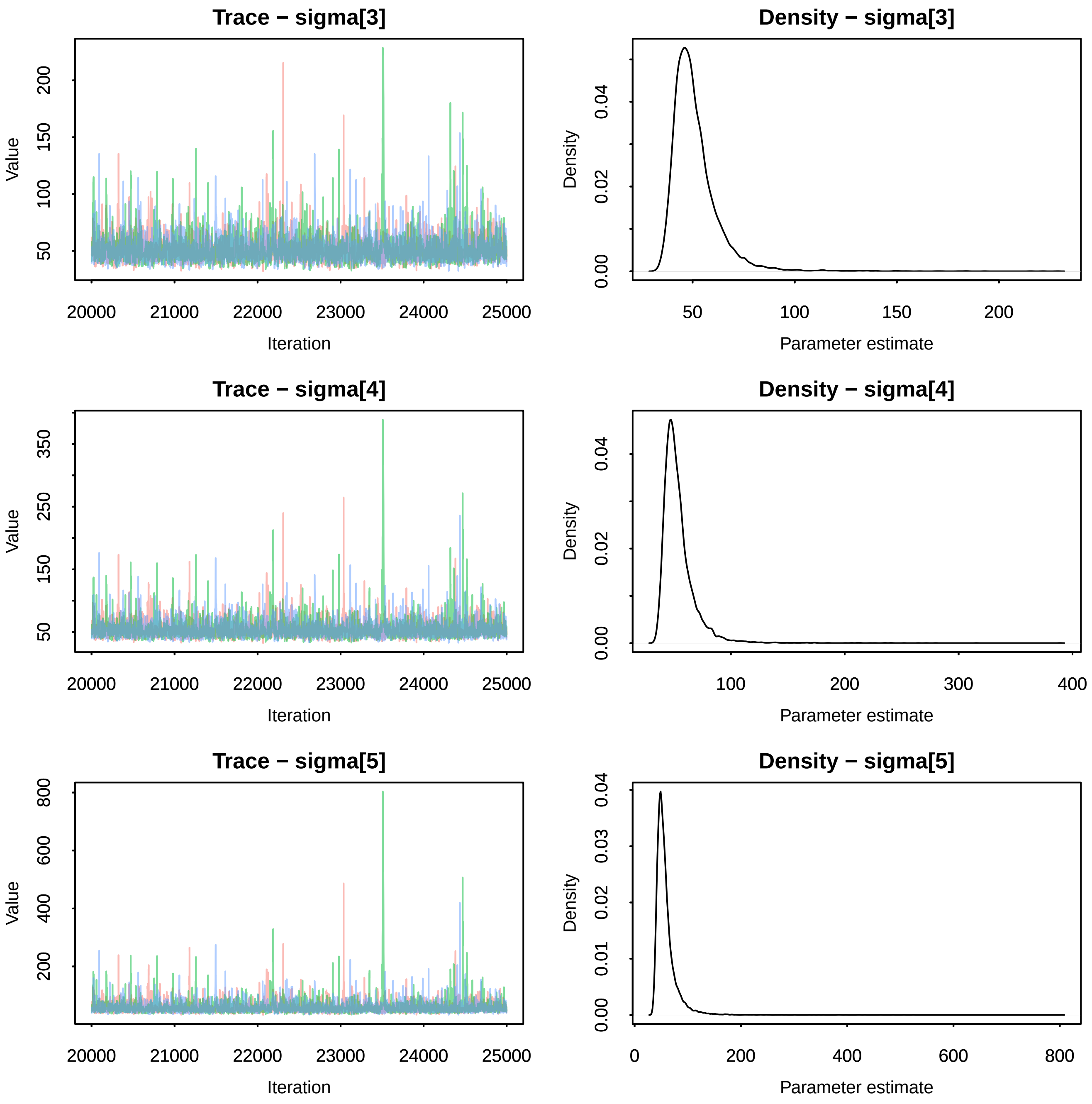


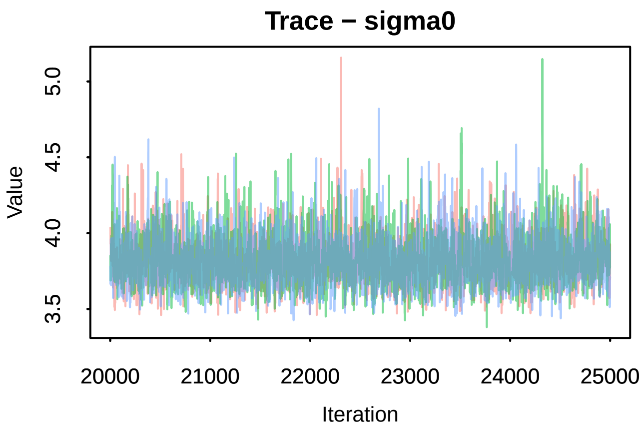
Trace − pi[1, 4]
Value
Iteration
Density − pi[1, 4]
Density
Parameter estimate
Trace − pi[2, 4]
Value
Iteration
Density − pi[2, 4]
Density
Parameter estimate
Trace − pi[3, 4]
Value
Iteration
Density − pi[3, 4]
Density
Parameter estimate

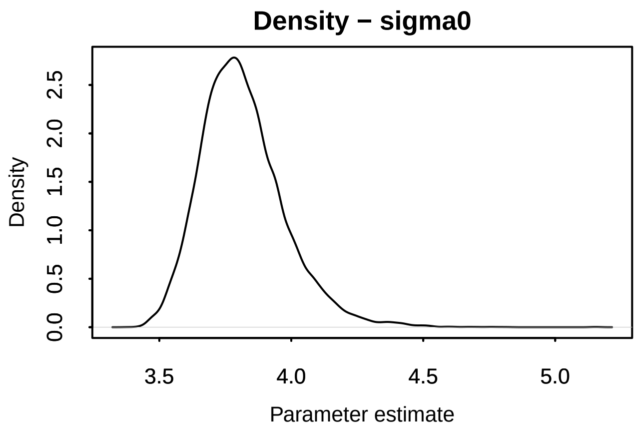

Trace − pi[4, 4]
Value
0.005
0.015
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[4, 4]
Density
0
50
100
150
0.000
0.005
0.010
0.015
0.020
Parameter estimate
Trace − pi[5, 4]
Value
0.000
0.005
0.010
0.015
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[5, 4]
Density
0
50
100
150
200
250
0.000
0.005
0.010
0.015
Parameter estimate
Trace − pi[1, 5]
Value
0.00
0.02
0.04
0.06
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[1, 5]
Density
0
20
40
60
80
0.00
0.02
0.04
0.06
Parameter estimate

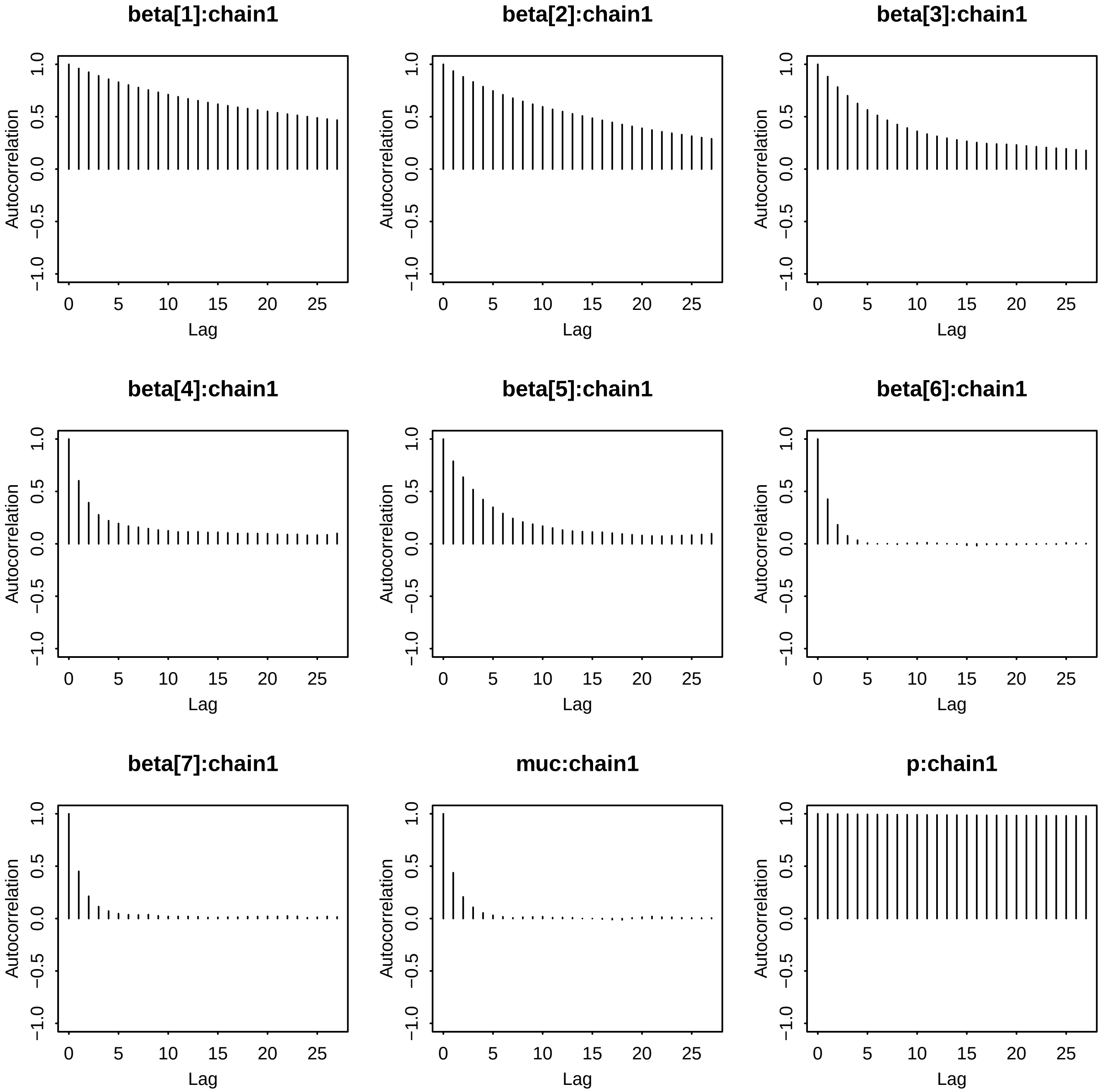

Trace − pi[2, 5]
Value
Iteration
Density − pi[2, 5]
Density
Parameter estimate
Trace − pi[3, 5]
Value
Iteration
Density − pi[3, 5]
Density
Parameter estimate
Trace − pi[4, 5]
Value
Iteration
Density − pi[4, 5]
Density
Parameter estimate

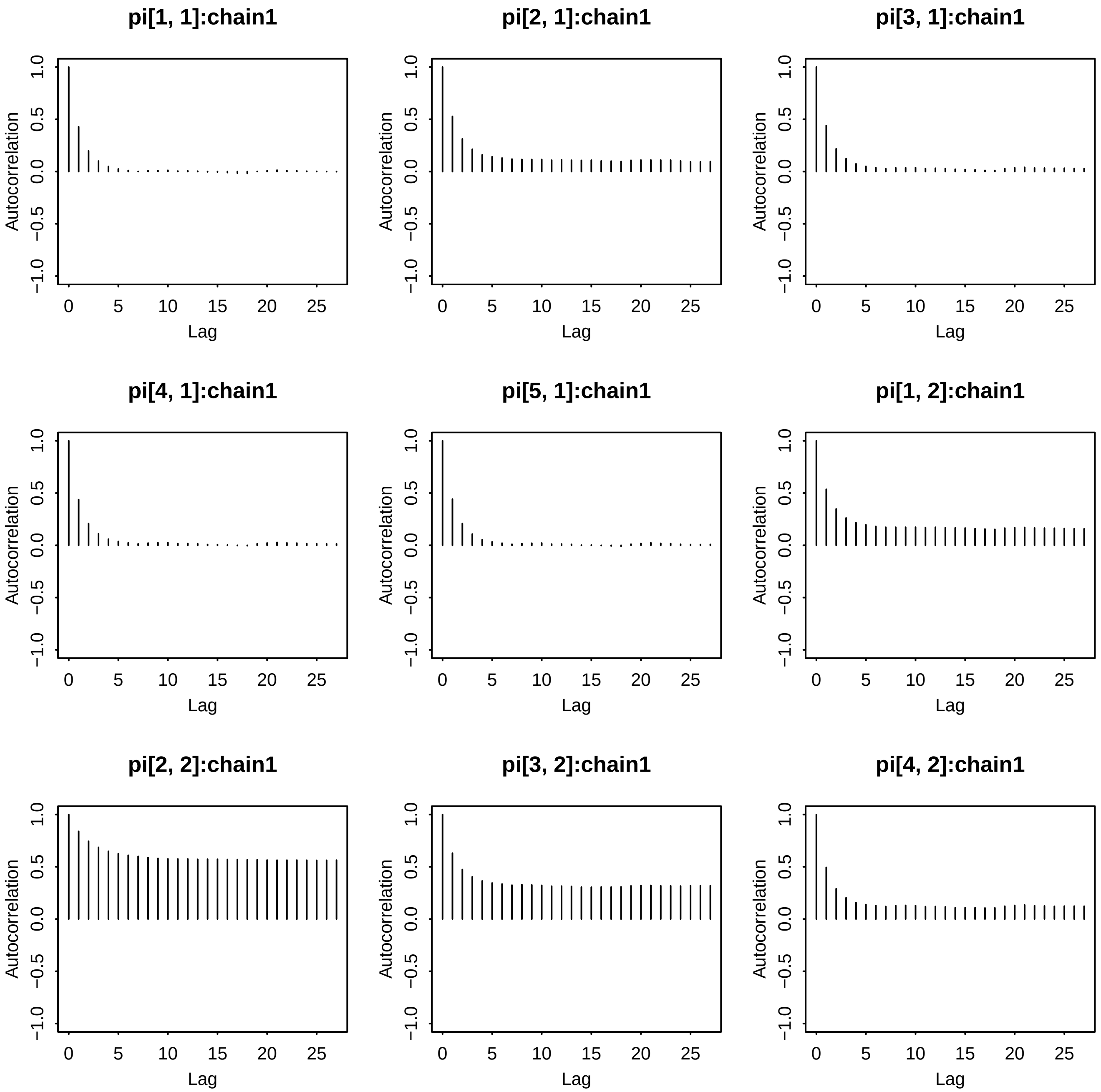

Trace − pi[5, 5]
Value
0.000
0.005
0.010
0.015
20000
21000
22000
23000
24000
25000
Iteration
Density − pi[5, 5]
Density
0
50
150
250
350
0.000
0.005
0.010
0.015
Parameter estimate
Trace − sigma[1]
Value
50
100
150
20000
21000
22000
23000
24000
25000
Iteration
Density − sigma[1]
Density
0.00
0.02
0.04
0.06
50
100
150
Parameter estimate
Trace − sigma[2]
Value
50
100
150
20000
21000
22000
23000
24000
25000
Iteration
Density − sigma[2]
Density
0.00
0.02
0.04
0.06
50
100
150
200
Parameter estimate

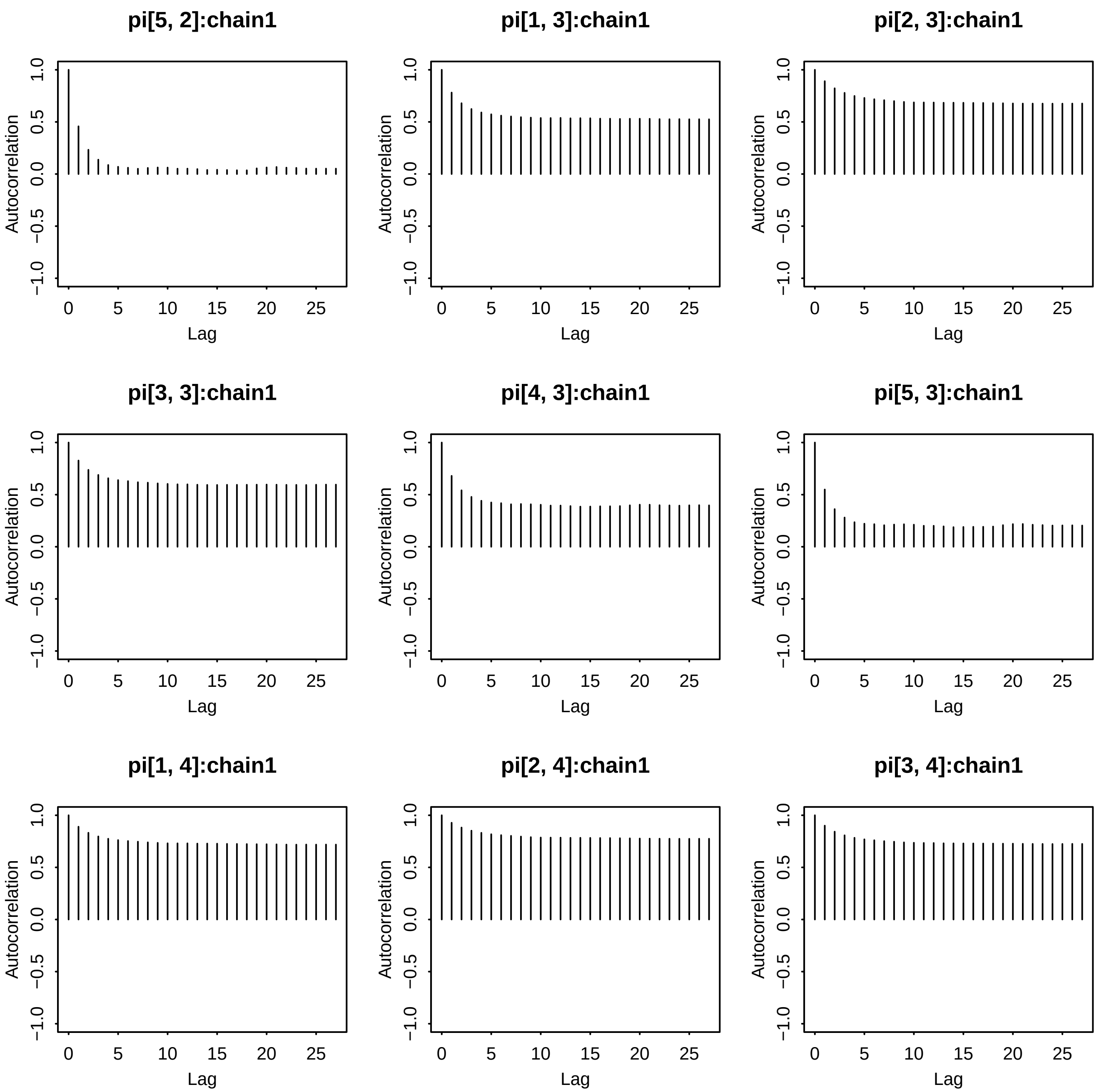

Trace − sigma[3]
Value
50
100
150
200
20000
21000
22000
23000
24000
25000
Iteration
Density − sigma[3]
Density
0.00
0.02
0.04
50
100
150
200
Parameter estimate
Trace − sigma[4]
Value
50
150
250
350
20000
21000
22000
23000
24000
25000
Iteration
Density − sigma[4]
Density
0.00
0.02
0.04
100
200
300
400
Parameter estimate
Trace − sigma[5]
Value
200
400
600
800
20000
21000
22000
23000
24000
25000
Iteration
Density − sigma[5]
Density
0.00
0.01
0.02
0.03
0.04
0
200
400
600
800
Parameter estimate

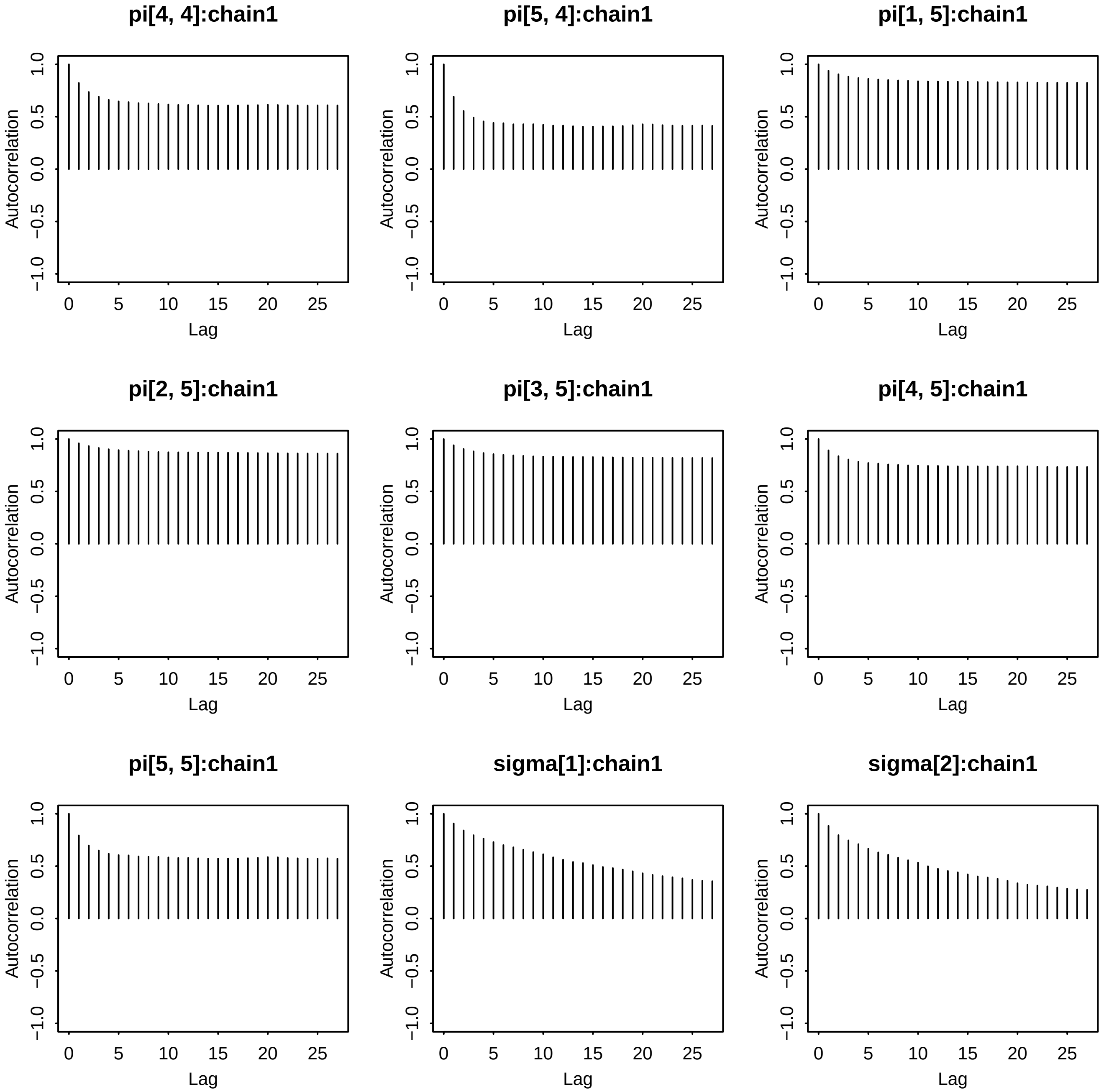

Trace − sigma0
Value
Iteration
3.5
4.0
4.5
5.0
20000
21000
22000
23000
24000
25000

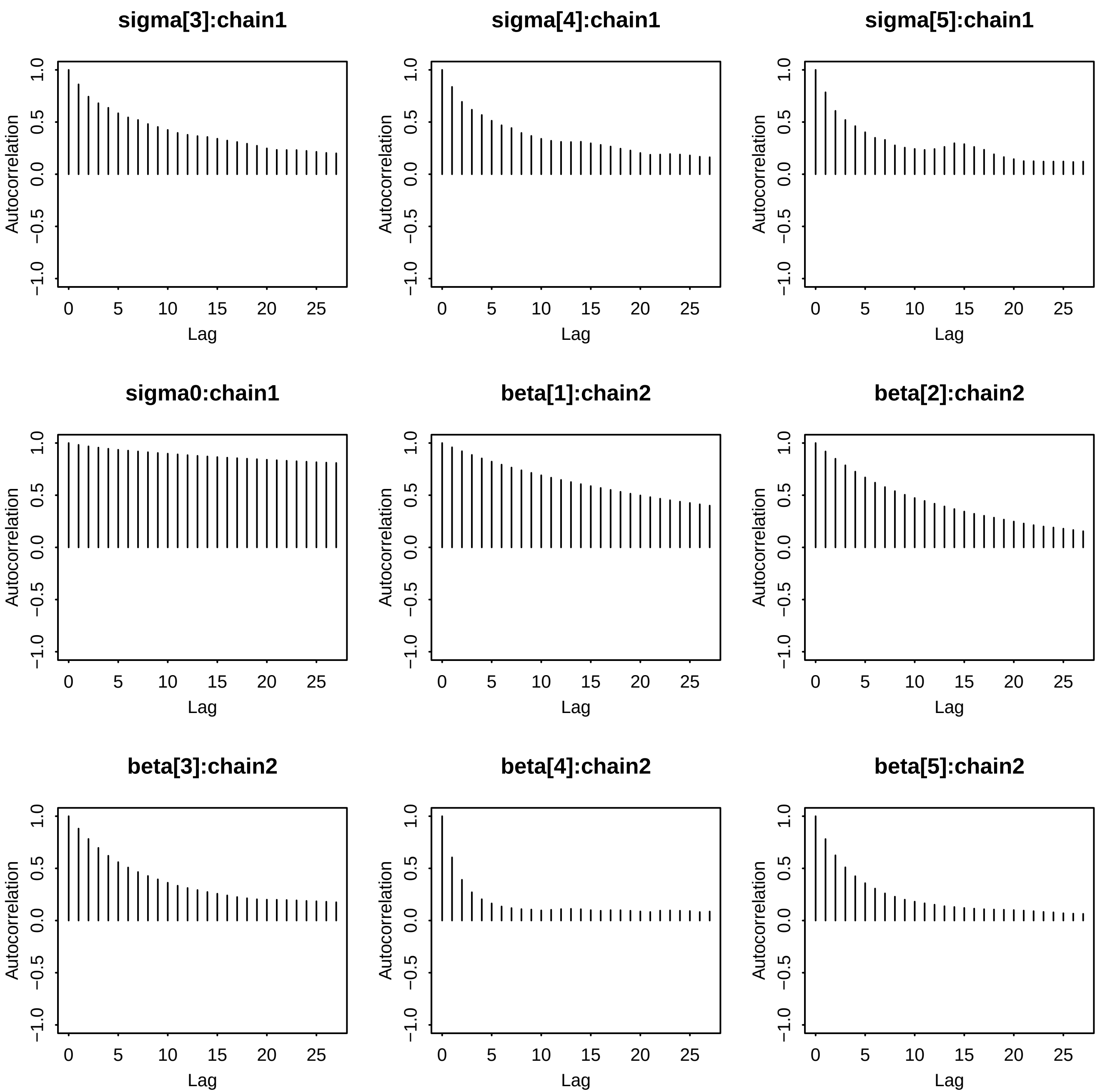

Density − sigma0
Density
Parameter estimate
0.0
0.5
1.0
1.5
2.0
2.5
3.5
4.0
4.5
5.0

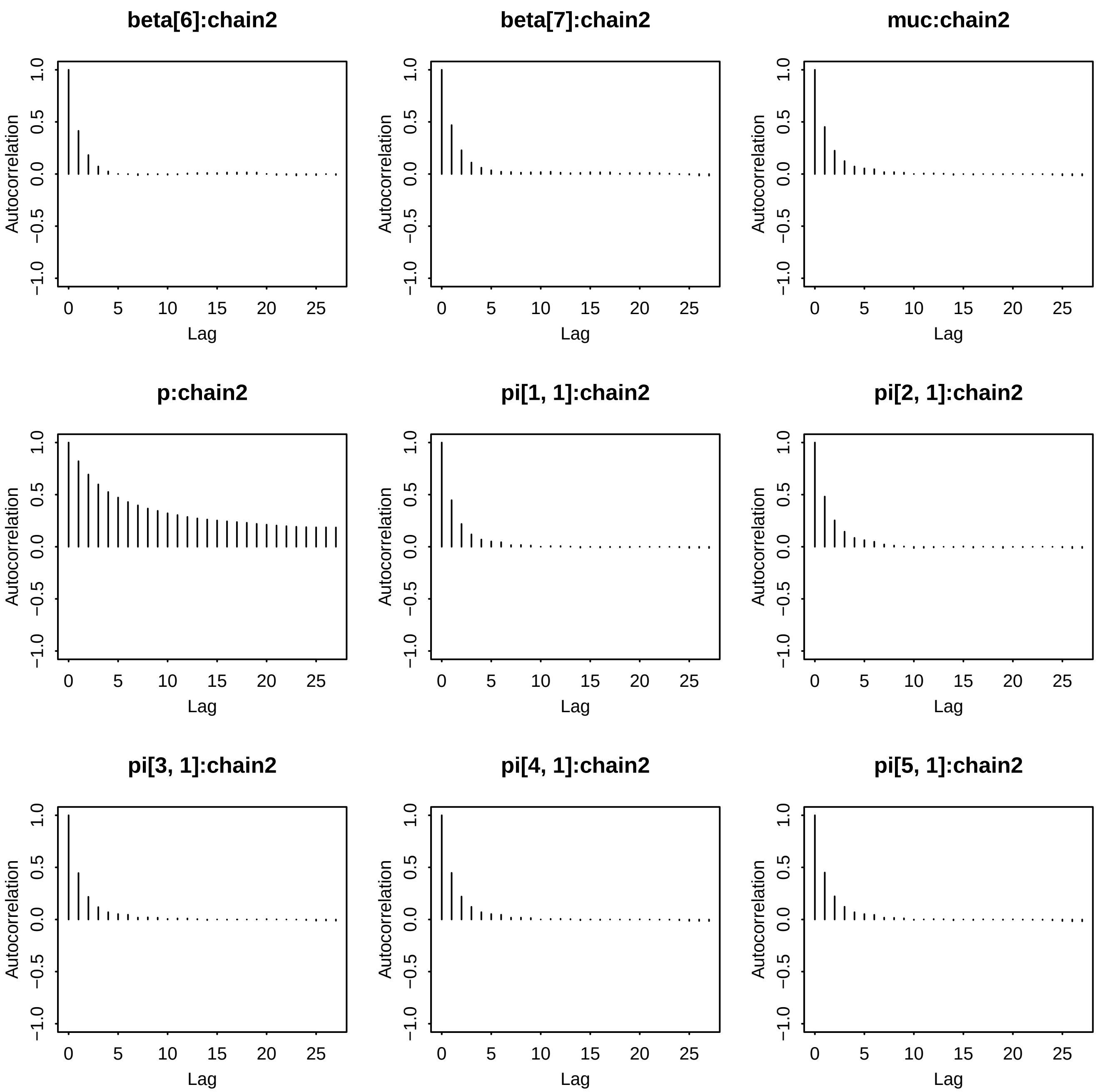

beta[1]:chain1
beta[2]:chain1
beta[3]:chain1
beta[4]:chain1
beta[5]:chain1
beta[6]:chain1
beta[7]:chain1
muc:chain1
p:chain1
Autocorrelation
Lag
−1.0
−0.5
0.0
0.5
1.0
0
5
10
15
20
25

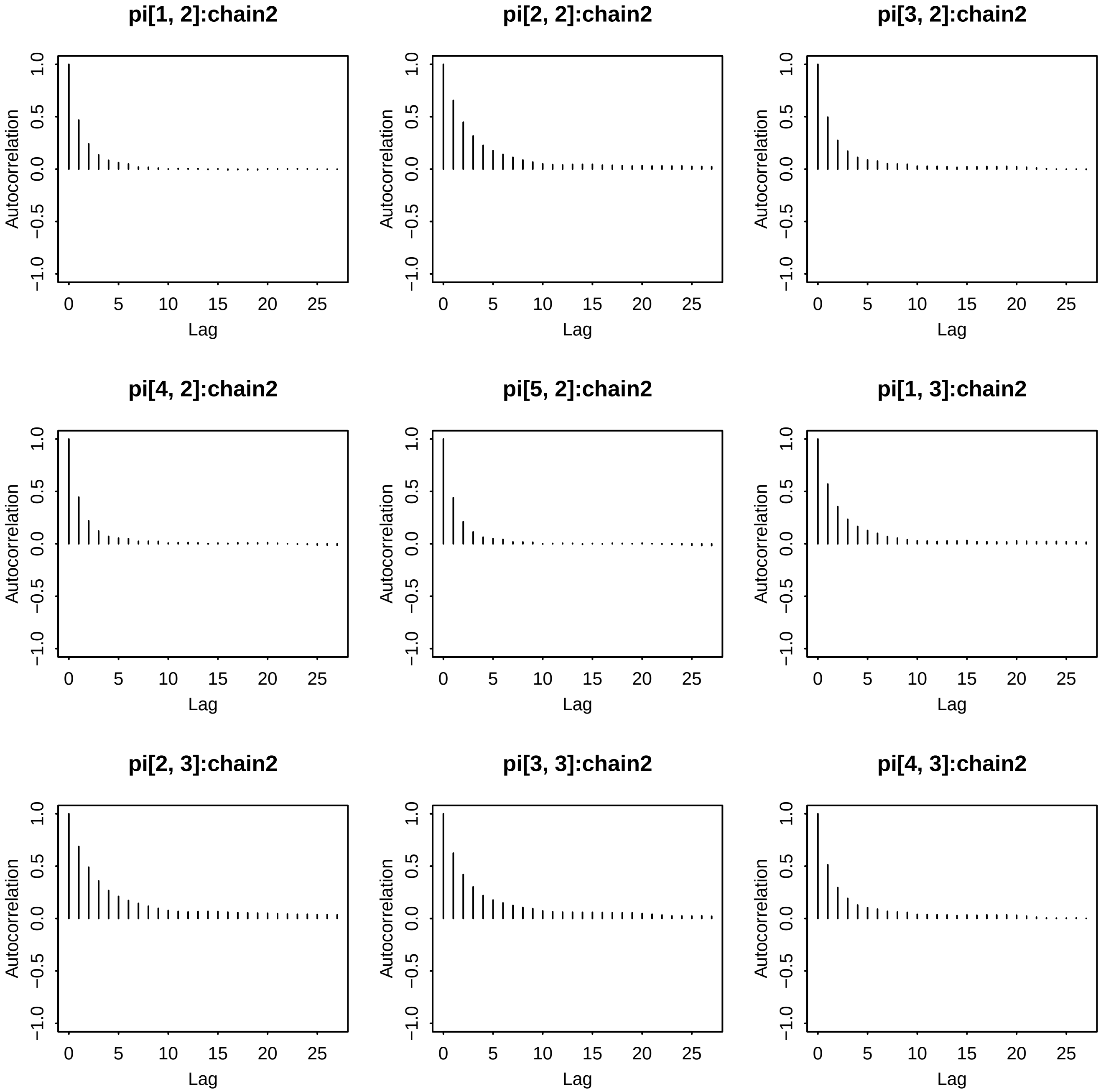

pi[1, 1]:chain1
pi[2, 1]:chain1
pi[3, 1]:chain1
pi[4, 1]:chain1
pi[5, 1]:chain1
pi[1, 2]:chain1
pi[2, 2]:chain1
pi[3, 2]:chain1
pi[4, 2]:chain1
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
Lag
0
5
10
15
20
25

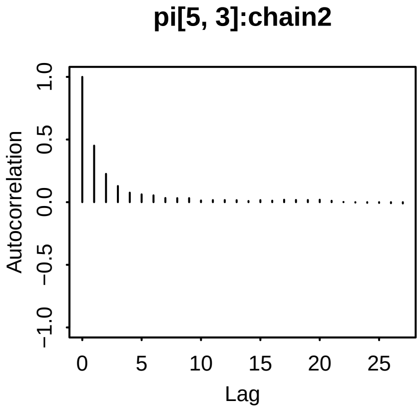

pi[5, 2]:chain1
pi[1, 3]:chain1
pi[2, 3]:chain1
pi[3, 3]:chain1
pi[4, 3]:chain1
pi[5, 3]:chain1
pi[1, 4]:chain1
pi[2, 4]:chain1
pi[3, 4]:chain1
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
Lag
0 5 10 15 20 25

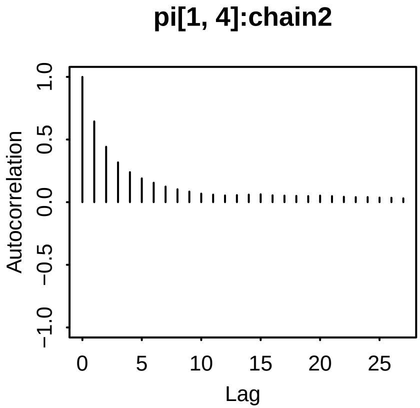

pi[4, 4]:chain1
pi[5, 4]:chain1
pi[1, 5]:chain1
pi[2, 5]:chain1
pi[3, 5]:chain1
pi[4, 5]:chain1
pi[5, 5]:chain1
sigma[1]:chain1
sigma[2]:chain1
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
Lag
0
5
10
15
20
25

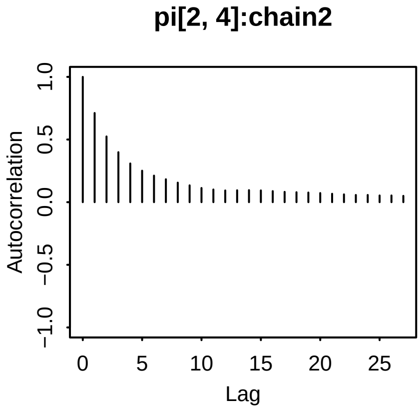

sigma[3]:chain1
sigma[4]:chain1
sigma[5]:chain1
sigma0:chain1
beta[1]:chain2
beta[2]:chain2
beta[3]:chain2
beta[4]:chain2
beta[5]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
Lag
0 5 10 15 20 25

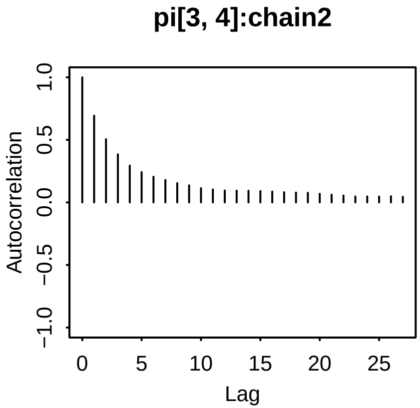

beta[6]:chain2
beta[7]:chain2
muc:chain2
p:chain2
pi[1, 1]:chain2
pi[2, 1]:chain2
pi[3, 1]:chain2
pi[4, 1]:chain2
pi[5, 1]:chain2
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
Lag
0
5
10
15
20
25

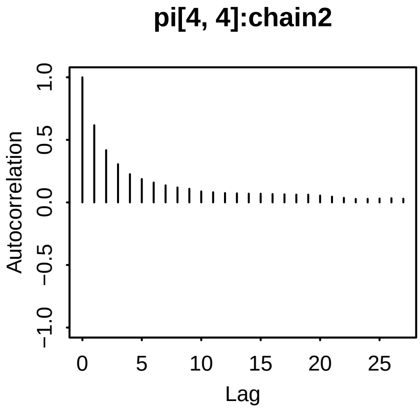

pi[1, 2]:chain2
pi[2, 2]:chain2
pi[3, 2]:chain2
pi[4, 2]:chain2
pi[5, 2]:chain2
pi[1, 3]:chain2
pi[2, 3]:chain2
pi[3, 3]:chain2
pi[4, 3]:chain2
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
Lag
0
5
10
15
20
25

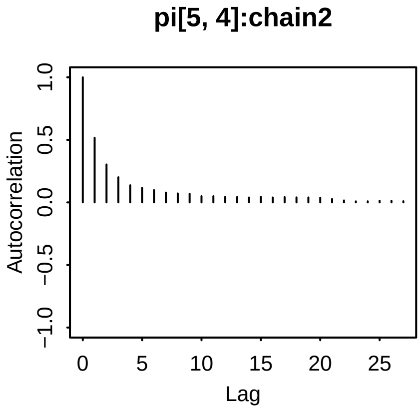

pi[5, 3]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

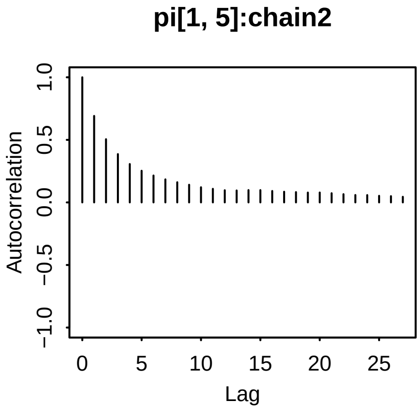

pi[1, 4]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

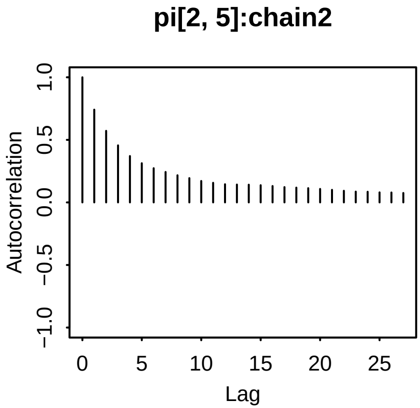

pi[2, 4]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

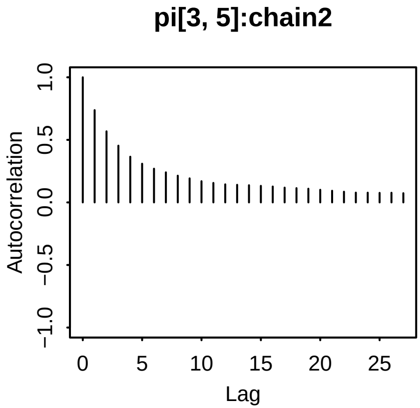

pi[3, 4]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

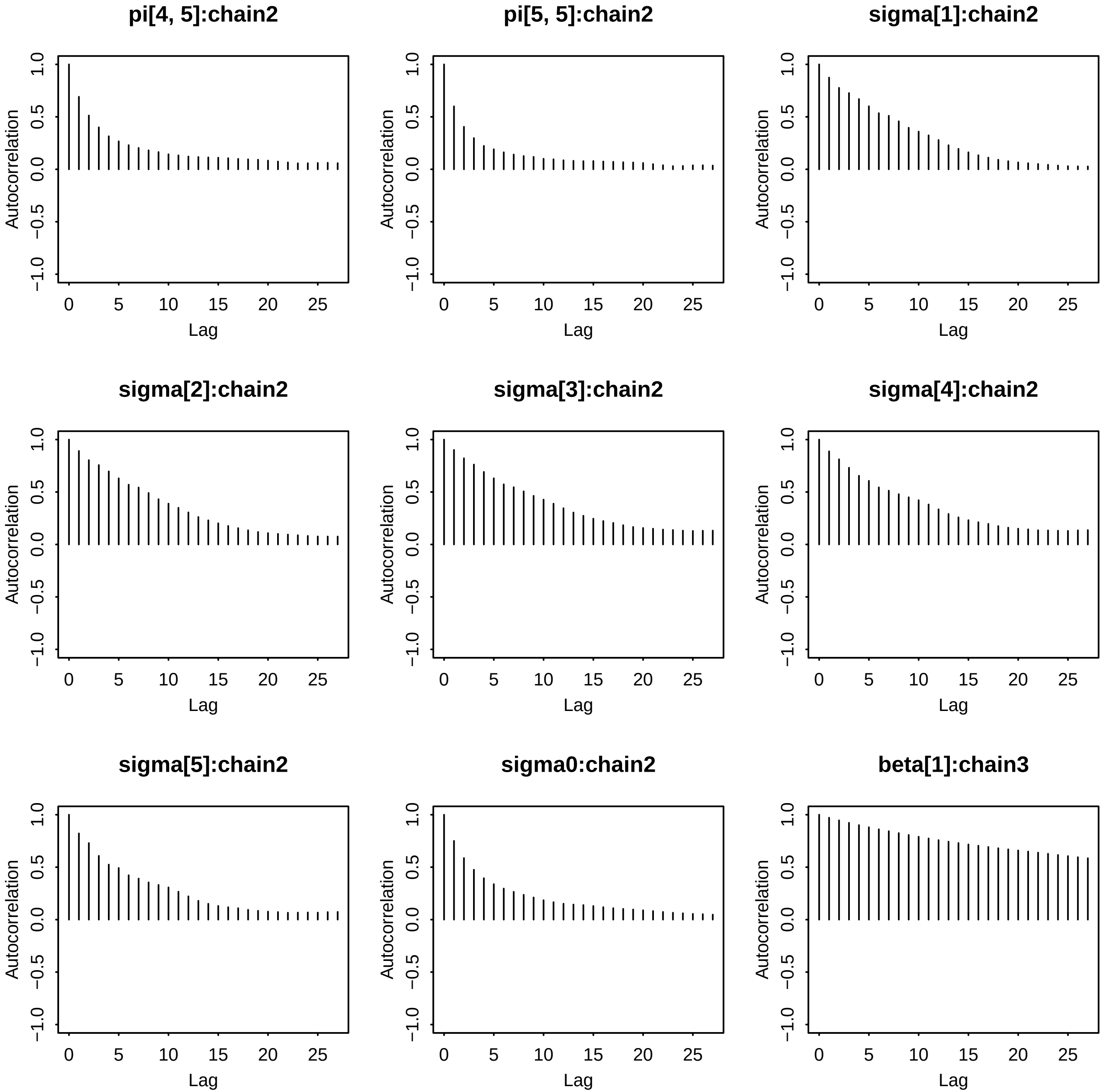

pi[4, 4]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

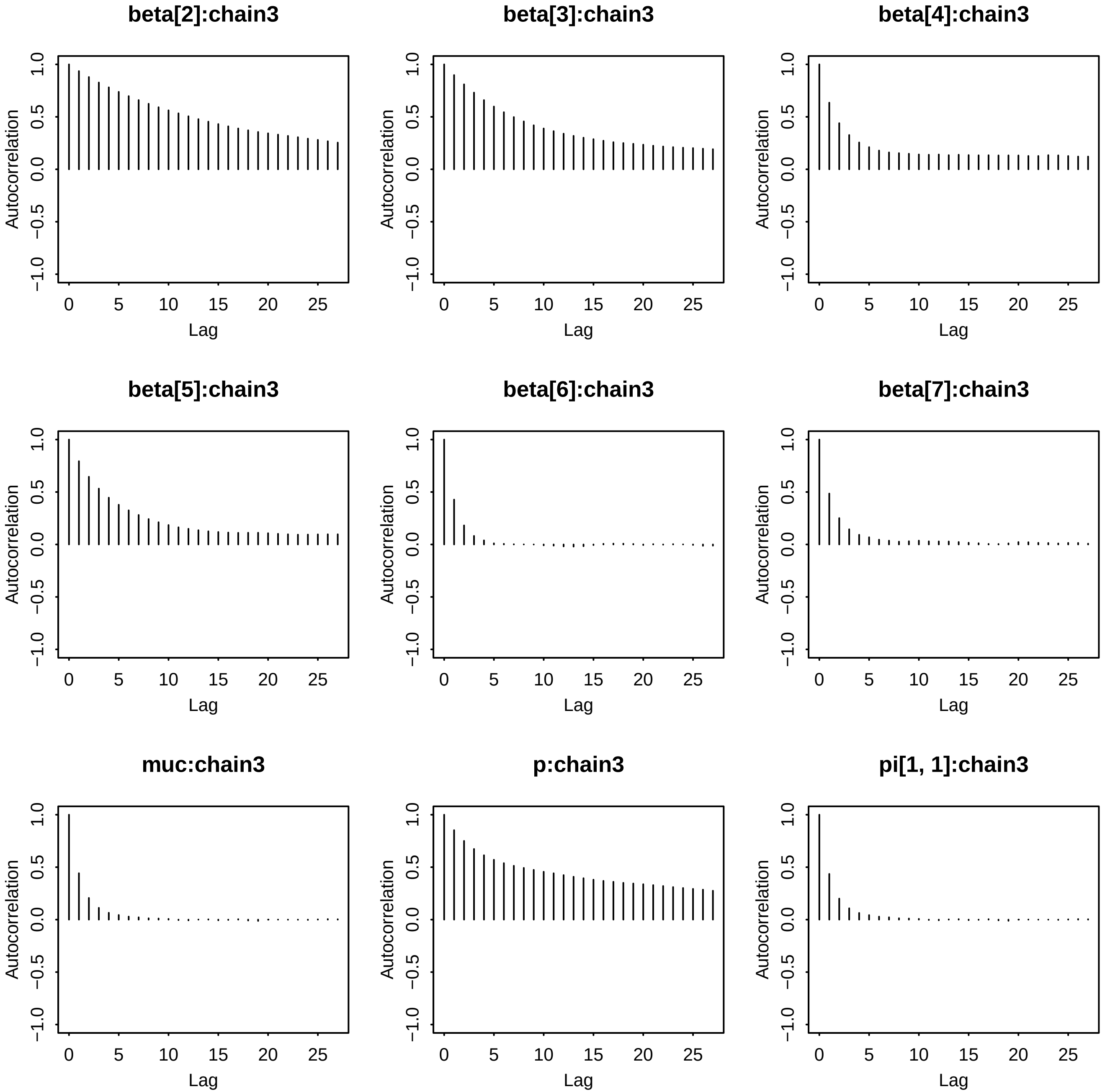

pi[5, 4]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

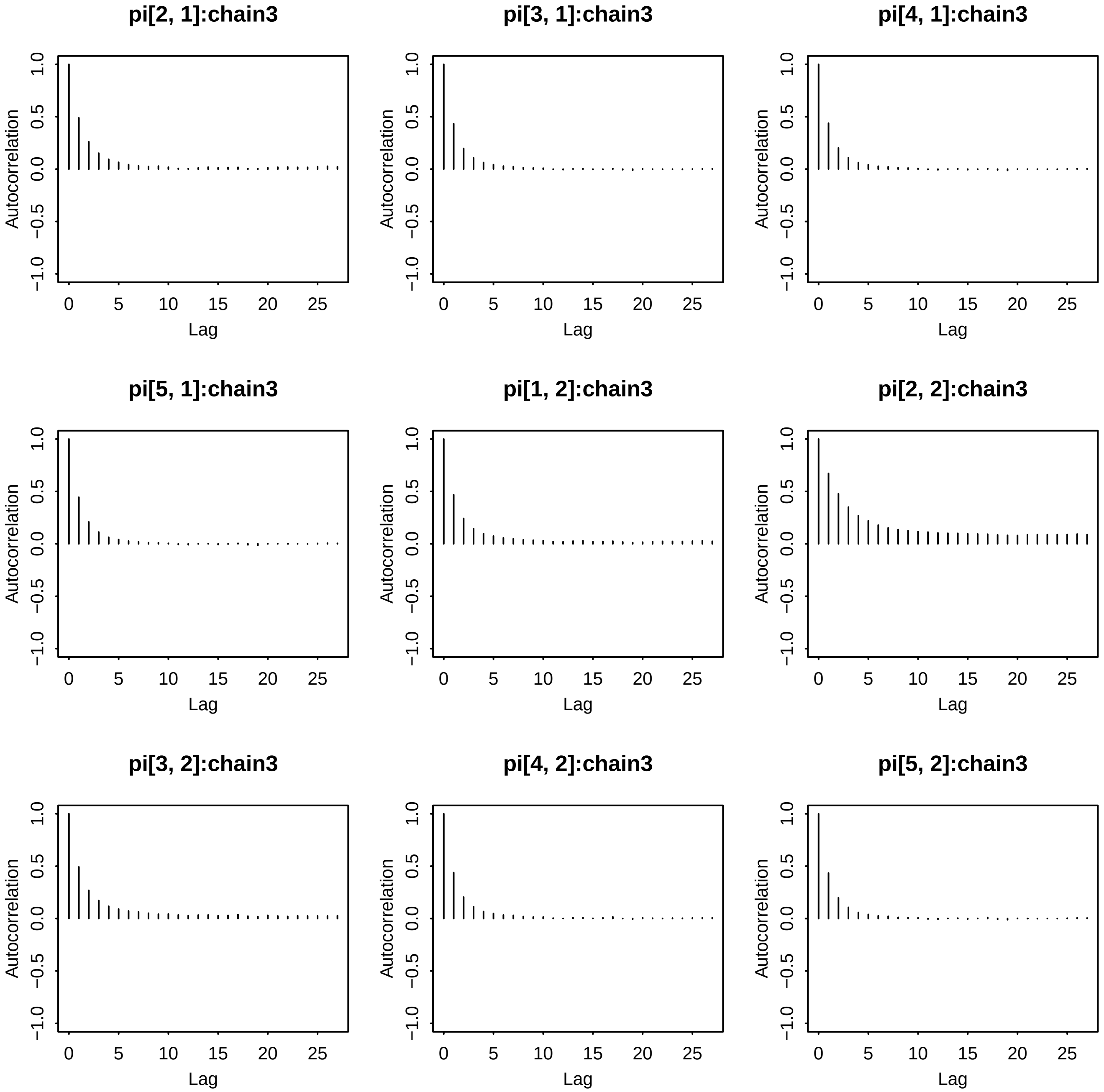

pi[1, 5]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

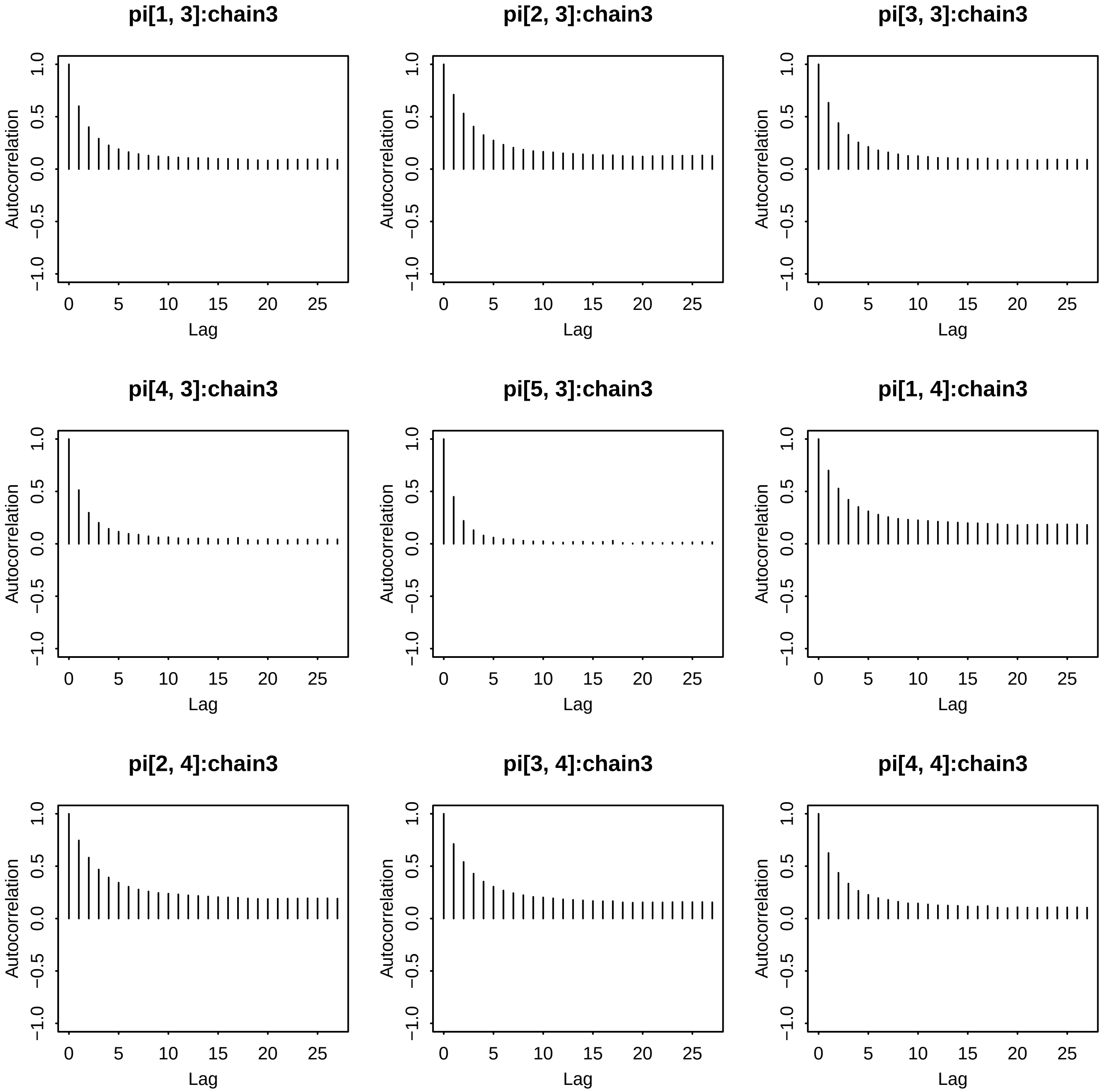

pi[2, 5]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

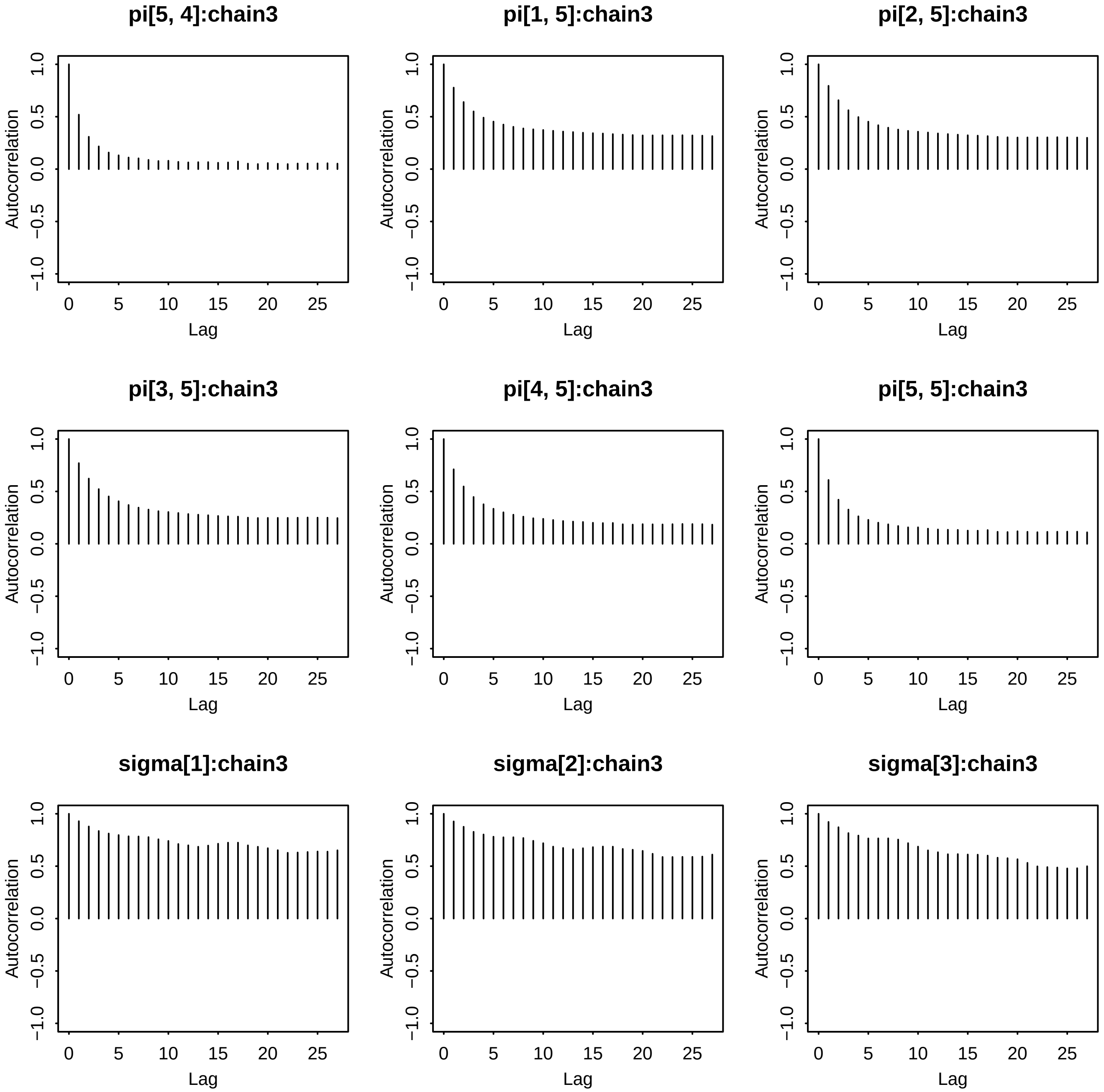

pi[3, 5]:chain2
Autocorrelation
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25
Lag

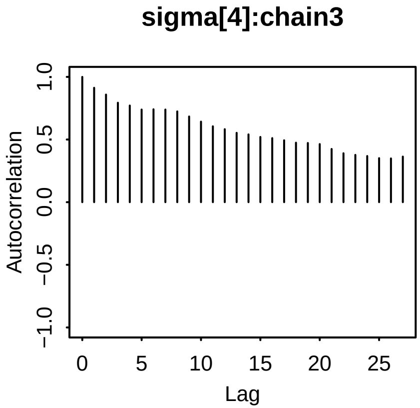

pi[4, 5]:chain2
Autocorrelation
Lag
pi[5, 5]:chain2
Autocorrelation
Lag
sigma[1]:chain2
Autocorrelation
Lag
sigma[2]:chain2
Autocorrelation
Lag
sigma[3]:chain2
Autocorrelation
Lag
sigma[4]:chain2
Autocorrelation
Lag
sigma[5]:chain2
Autocorrelation
Lag
sigma0:chain2
Autocorrelation
Lag
beta[1]:chain3
Autocorrelation
Lag
−1.0 −0.5 0.0 0.5 1.0
0 5 10 15 20 25

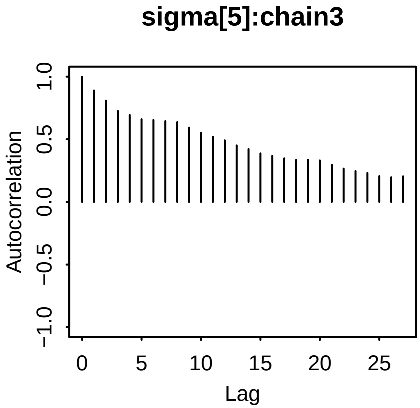

beta[2]:chain3
beta[3]:chain3
beta[4]:chain3
beta[5]:chain3
beta[6]:chain3
beta[7]:chain3
muc:chain3
p:chain3
pi[1, 1]:chain3
Autocorrelation
Lag
-1.0
-0.5
0.0
0.5
1.0
0
5
10
15
20
25

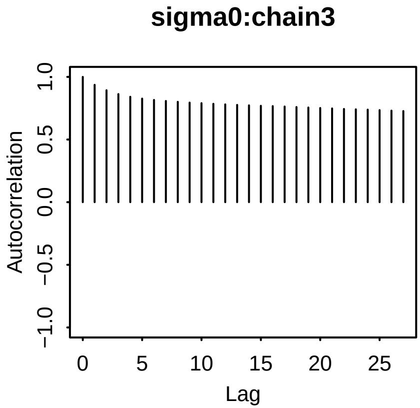

pi[2, 1]:chain3
pi[3, 1]:chain3
pi[4, 1]:chain3
pi[5, 1]:chain3
pi[1, 2]:chain3
pi[2, 2]:chain3
pi[3, 2]:chain3
pi[4, 2]:chain3
pi[5, 2]:chain3
Autocorrelation
Lag
−1.0
−0.5
0.0
0.5
1.0
0
5
10
15
20
25

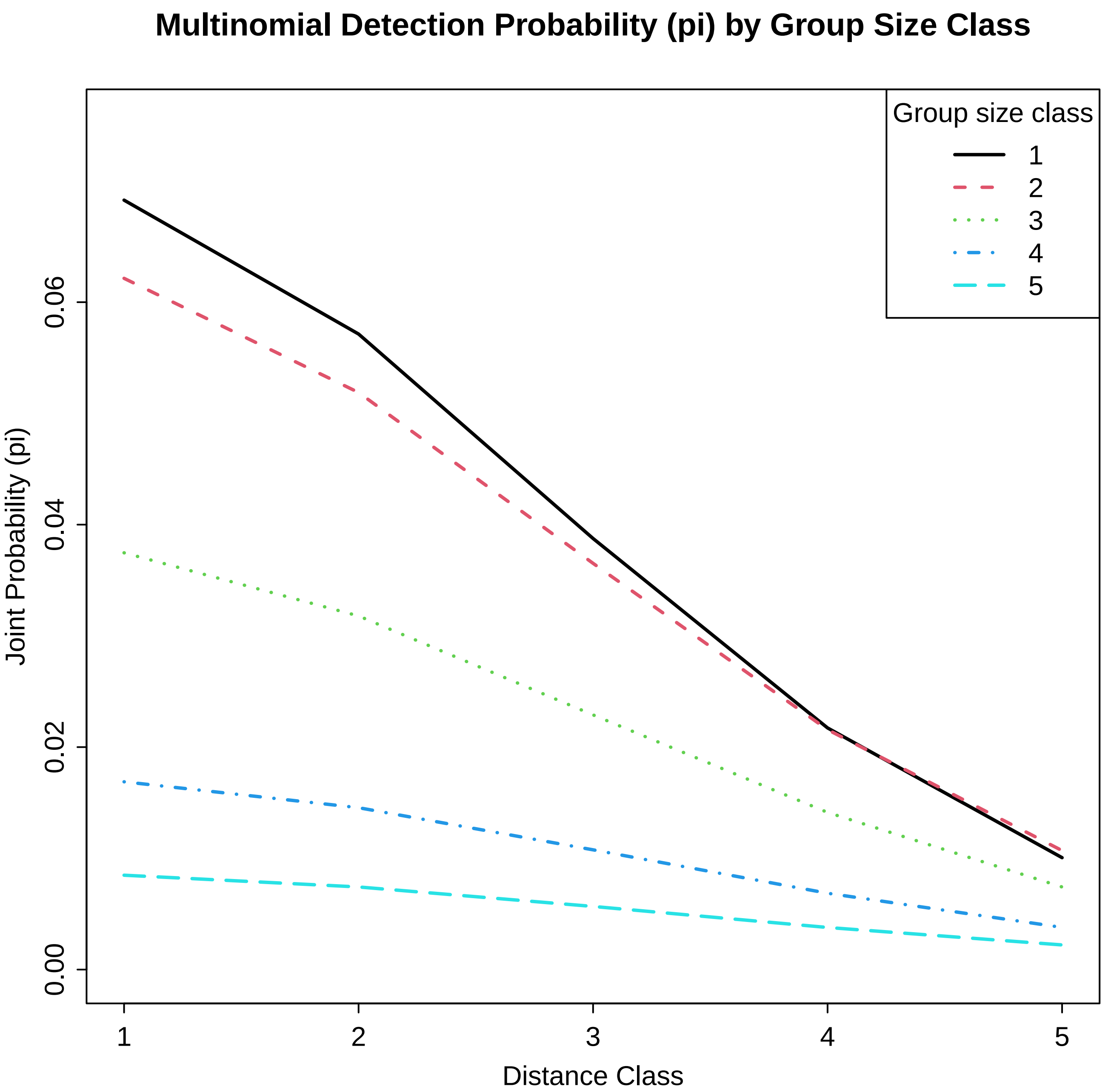

pi[1, 3]:chain3
pi[2, 3]:chain3
pi[3, 3]:chain3
pi[4, 3]:chain3
pi[5, 3]:chain3
pi[1, 4]:chain3
pi[2, 4]:chain3
pi[3, 4]:chain3
pi[4, 4]:chain3
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
Lag
0
5
10
15
20
25


pi[5, 4]:chain3
pi[1, 5]:chain3
pi[2, 5]:chain3
pi[3, 5]:chain3
pi[4, 5]:chain3
pi[5, 5]:chain3
sigma[1]:chain3
sigma[2]:chain3
sigma[3]:chain3
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
Lag
0
5
10
15
20
25

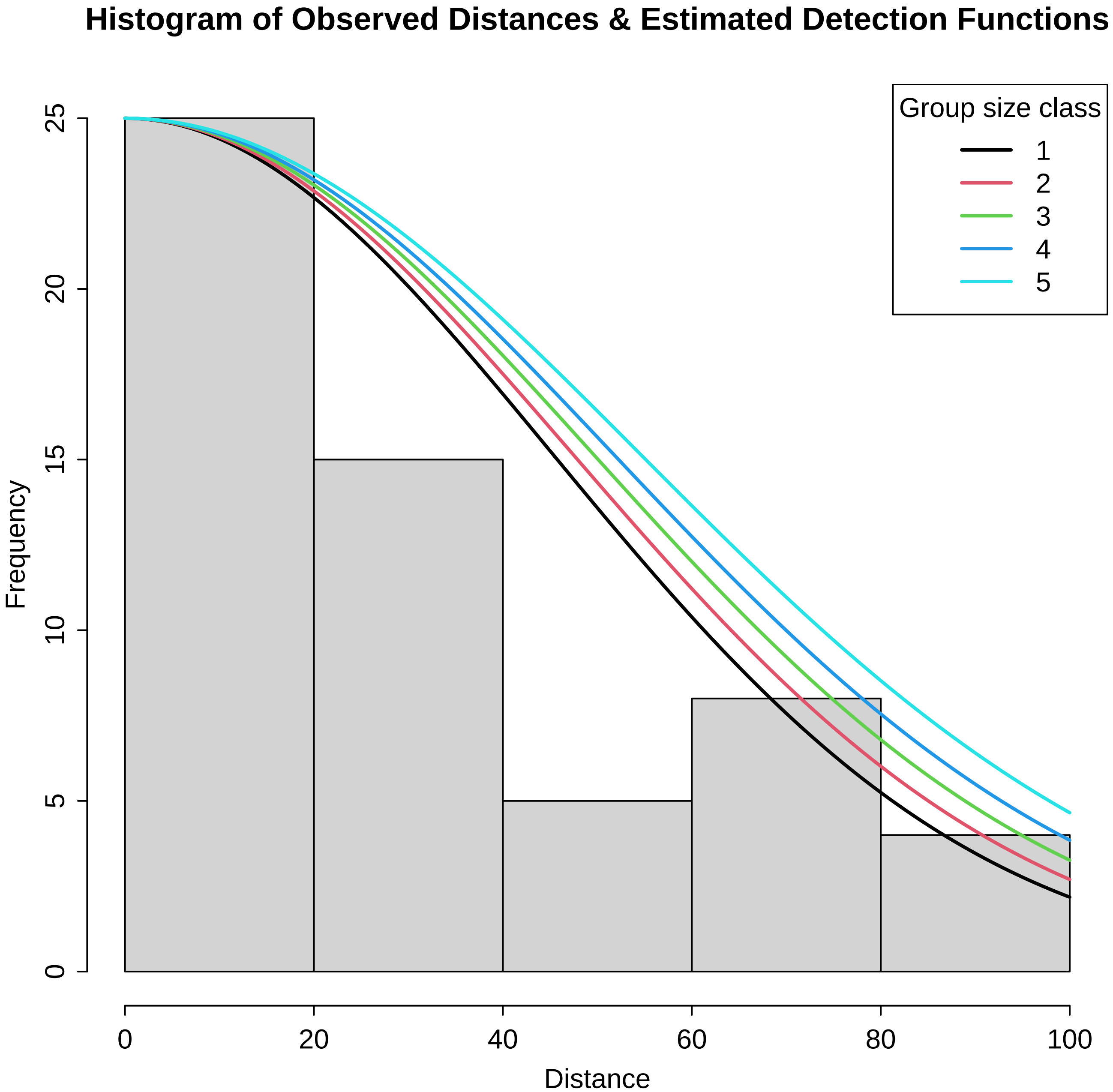

sigma[4]:chain3
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
0
5
10
15
20
25
Lag

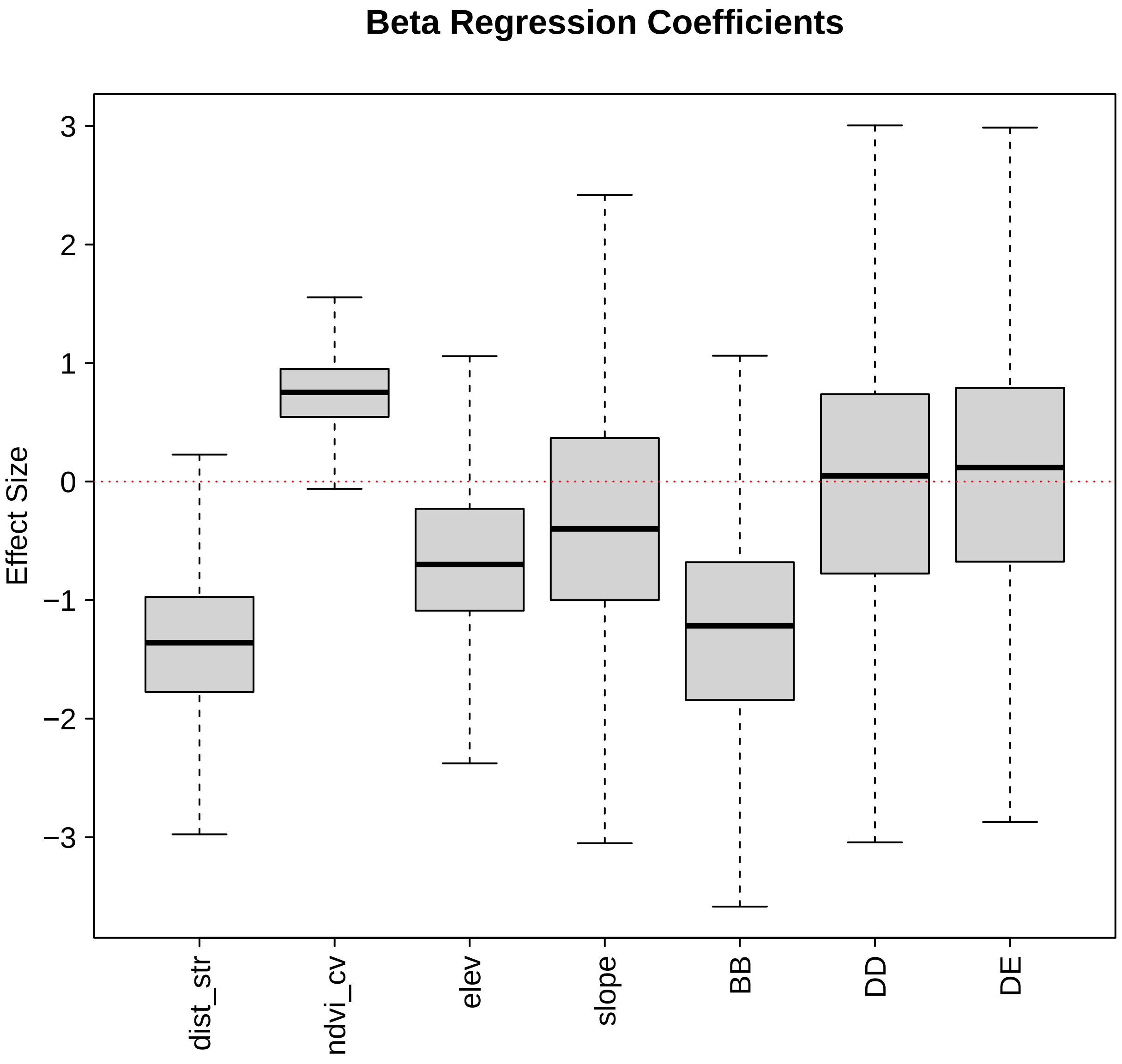

sigma[5]:chain3
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
0
5
10
15
20
25
Lag

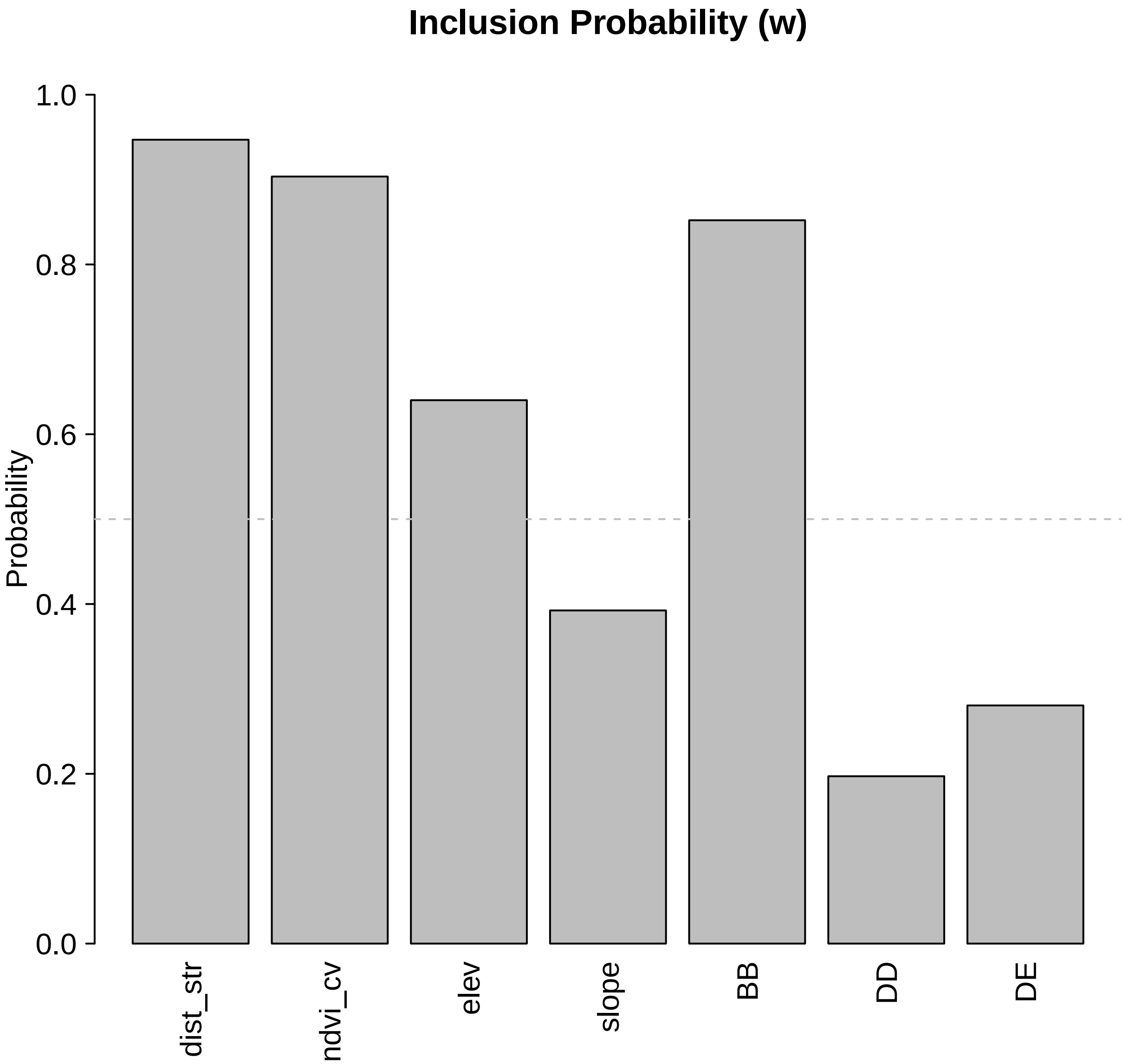

sigma0:chain3
Autocorrelation
−1.0
−0.5
0.0
0.5
1.0
0
5
10
15
20
25
Lag

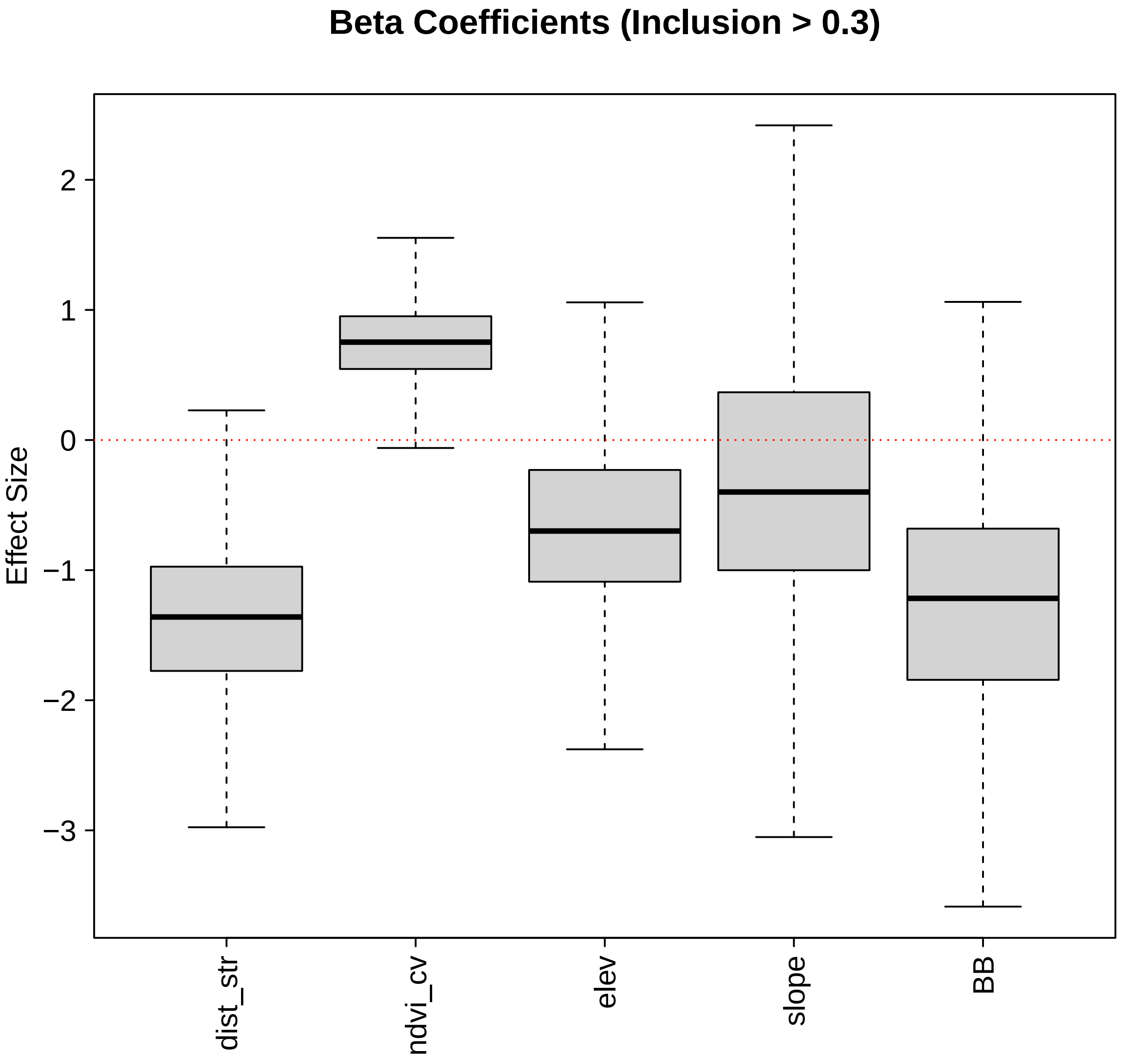


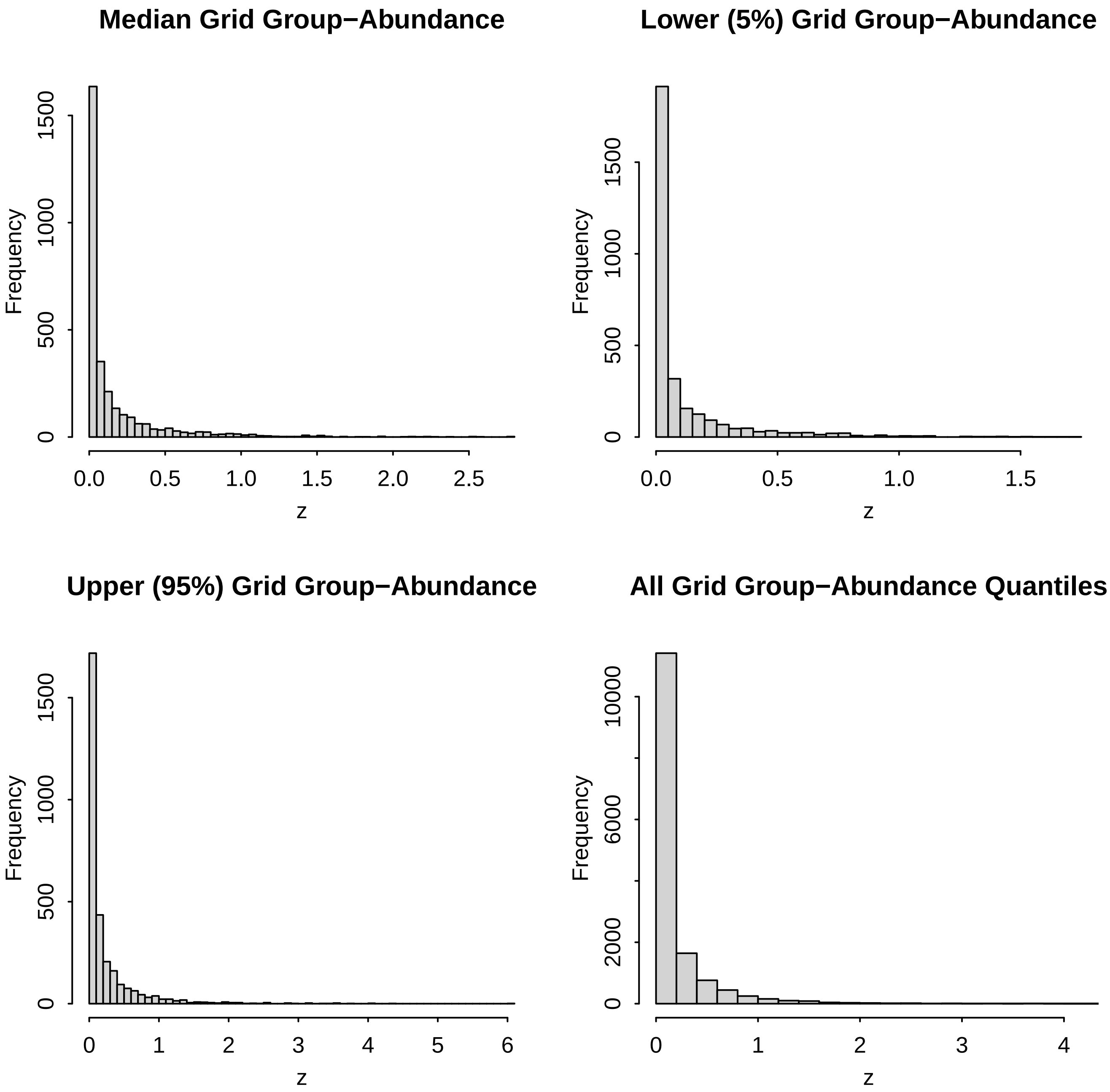
Multinomial Detection Probability (pi) by Group Size Class
Group size class
1
2
3
4
5
Joint Probability (pi)
0.00
0.02
0.04
0.06
Distance Class
1
2
3
4
5

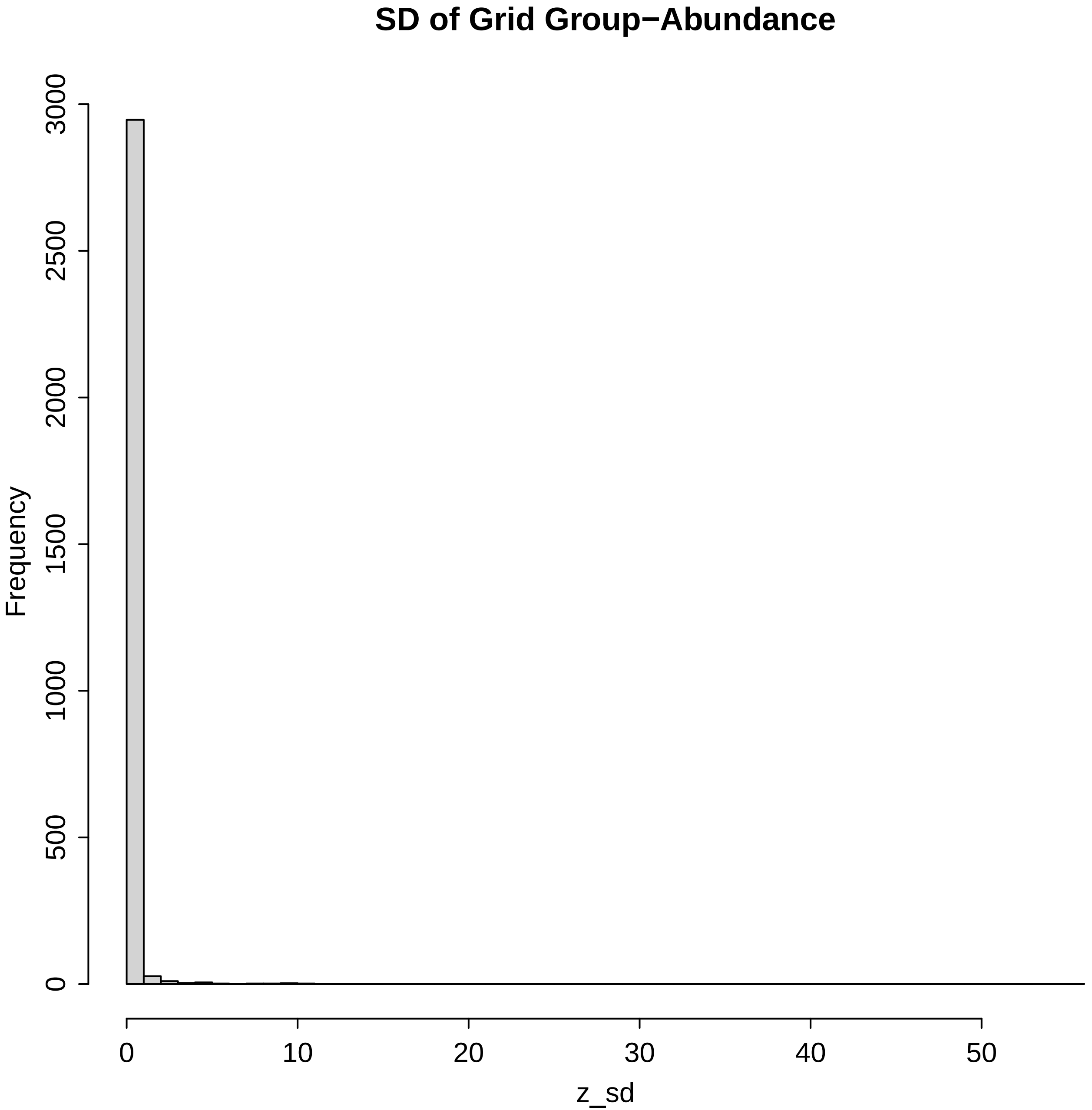

Histogram of Observed Distances & Estimated Detection Functions
Group size class
1
2
3
4
5
Frequency
0
5
10
15
20
25
Distance
0
20
40
60
80
100


Beta Regression Coefficients
Effect Size
3
2
1
0
−1
−2
−3
dist_str
ndvi_cv
elev
slope
BB
DD
DE

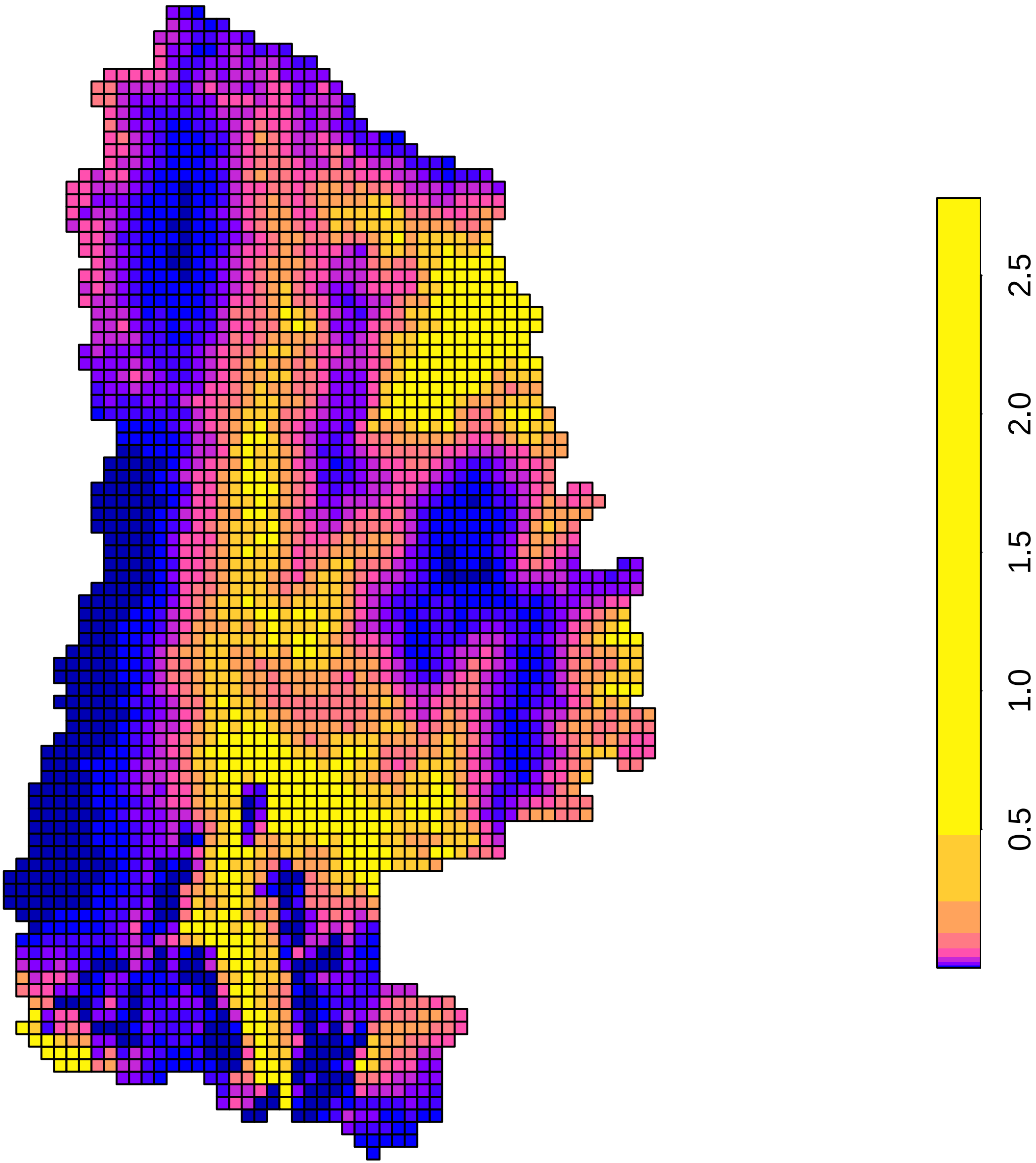

Inclusion Probability (w)
Probability
1.0
0.8
0.6
0.4
0.2
0.0
dist_str
ndvi_cv
elev
slope
BB
DD
DE


Beta Coefficients (Inclusion > 0.3)
Effect Size
2
1
0
−1
−2
−3
dist_str
ndvi_cv
elev
slope
BB

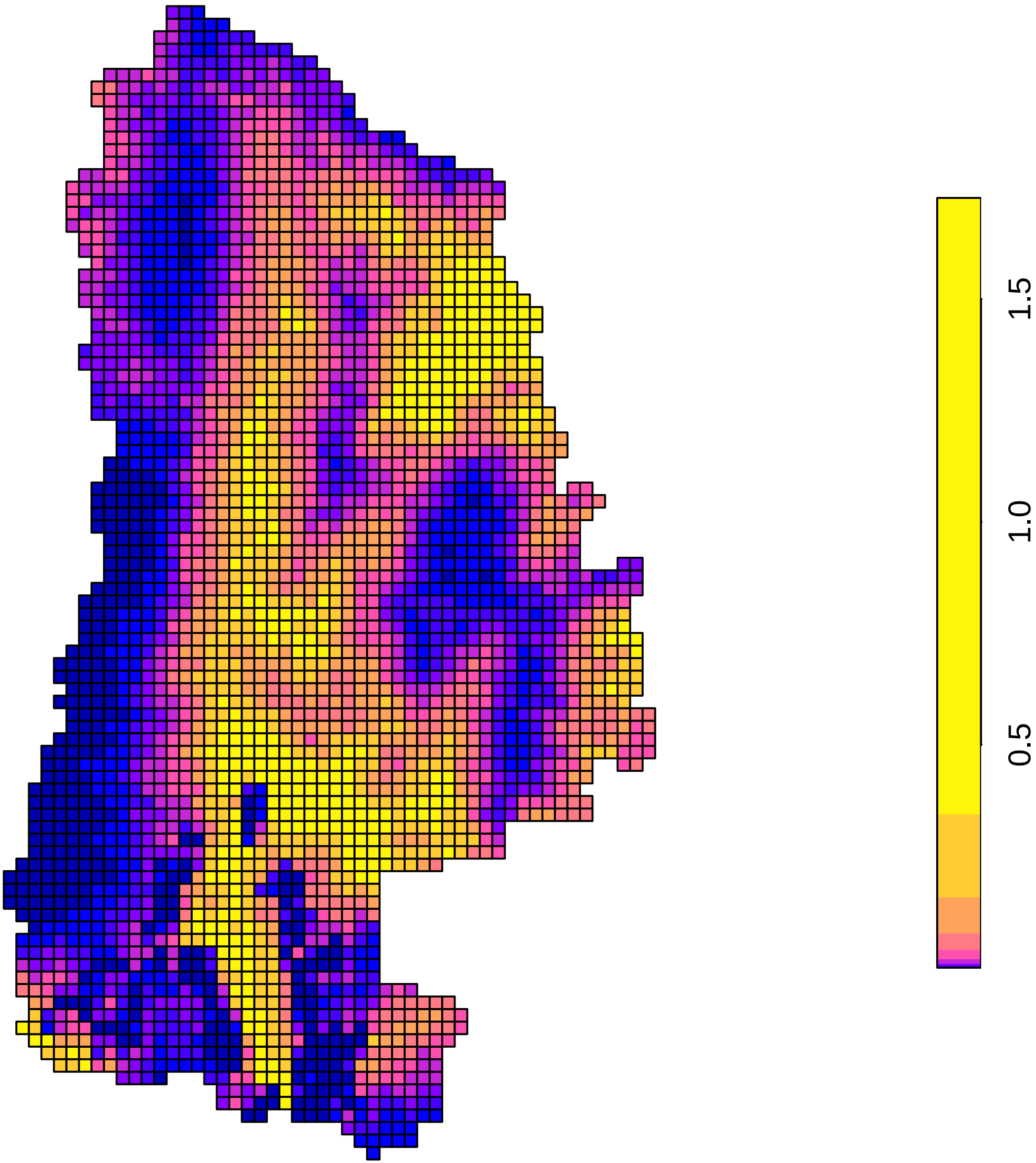

Median Grid Group−Abundance
Frequency
z
Lower (5%) Grid Group−Abundance
Frequency
z
Upper (95%) Grid Group−Abundance
Frequency
z
All Grid Group−Abundance Quantiles
Frequency
z


SD of Grid Group−Abundance
Frequency
0
500
1000
1500
2000
2500
3000
0
10
20
30
40
50
z_sd

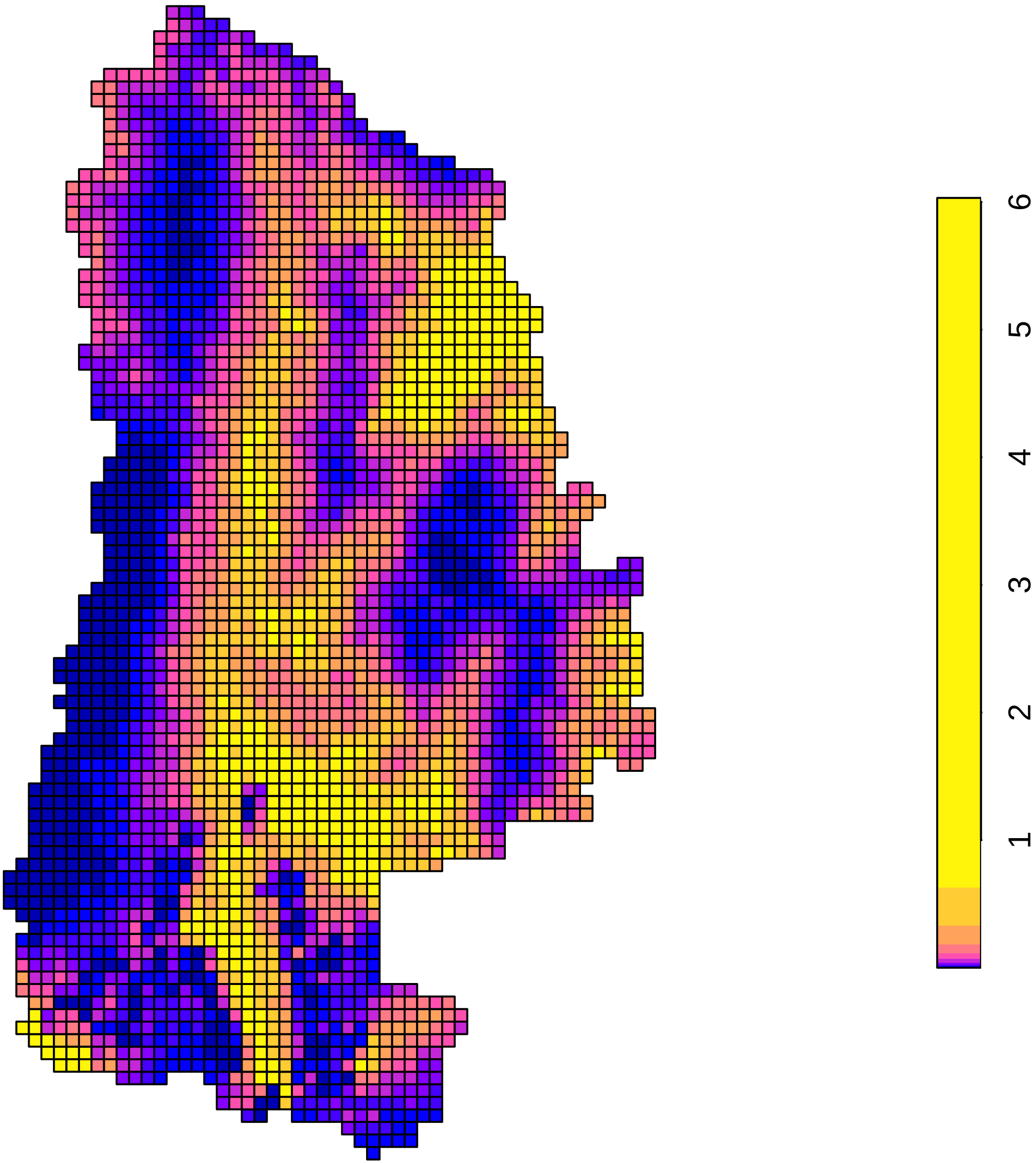
# Median Grid Group Abundance (z)



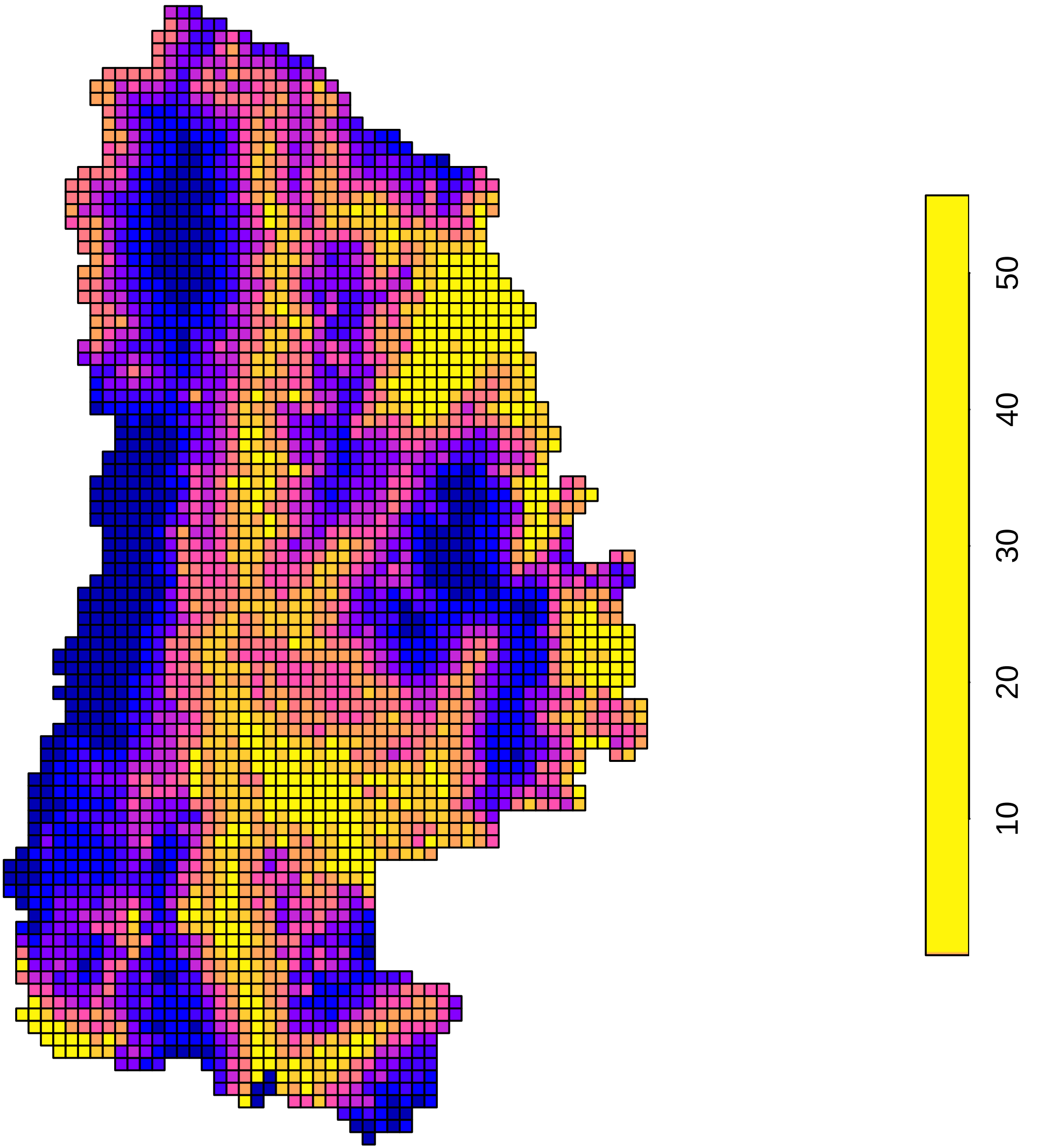
Lower (5%) Grid Group Abundance (z)

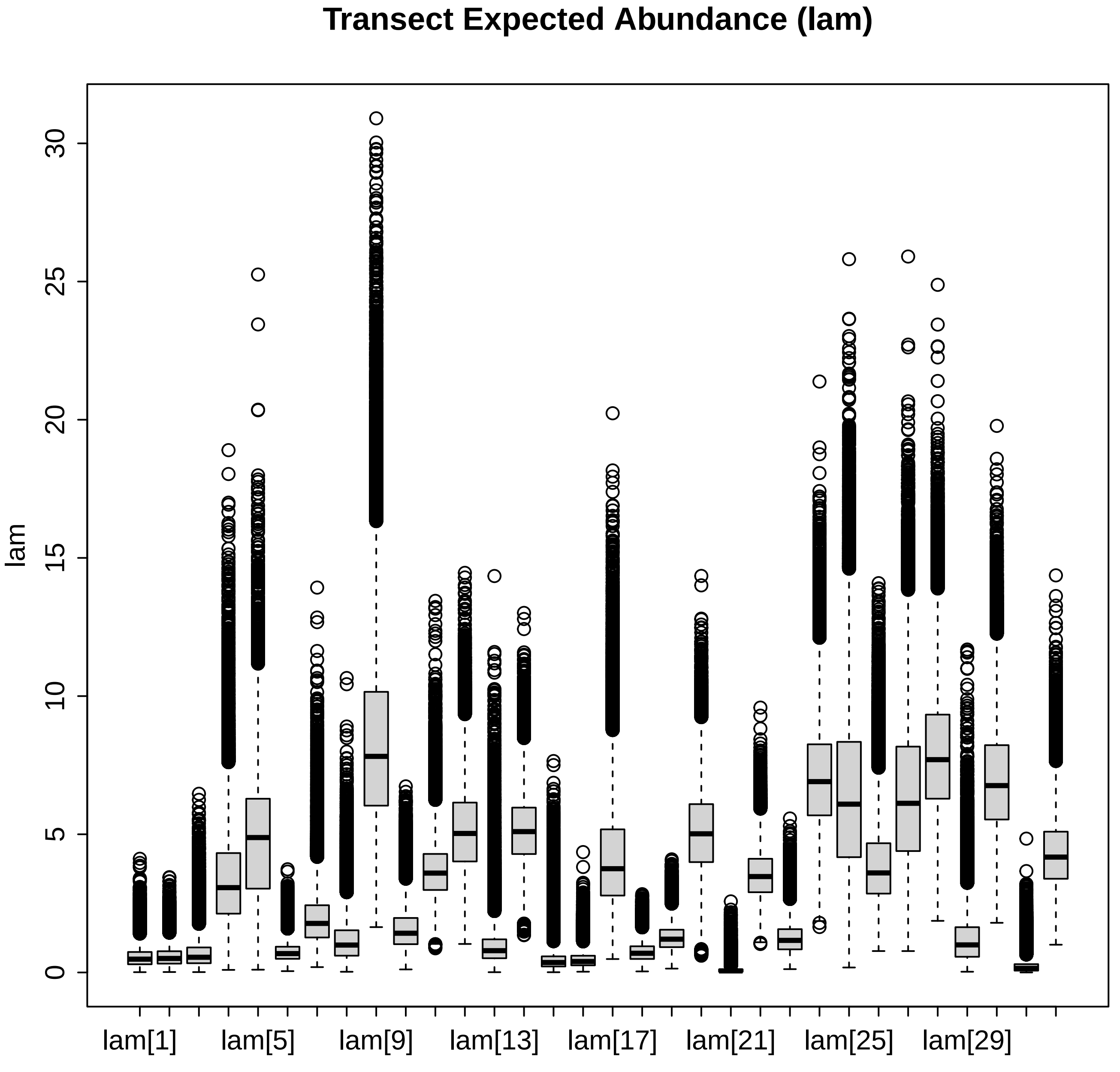


# Upper (95%) Grid Group Abundance (z)

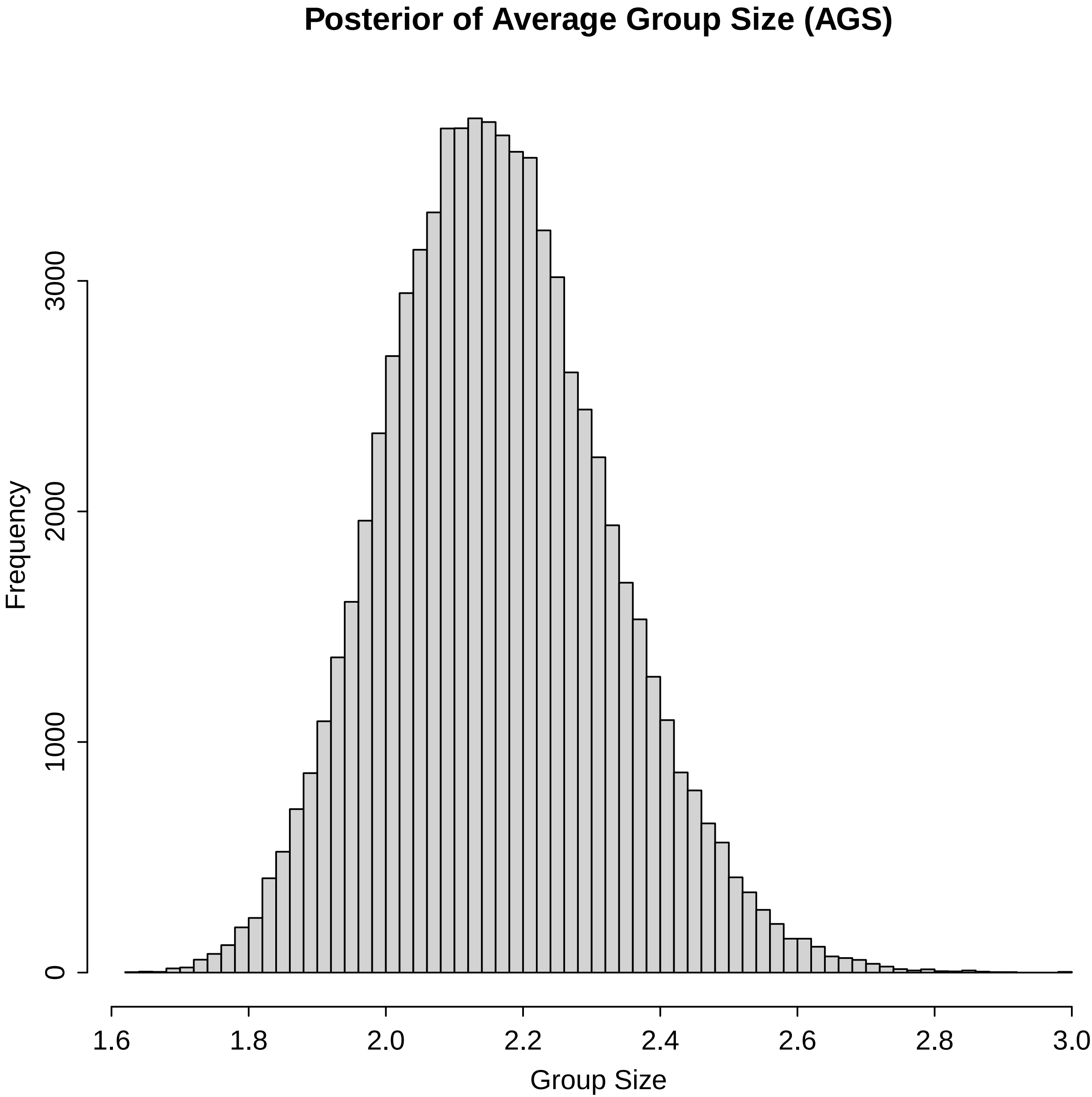


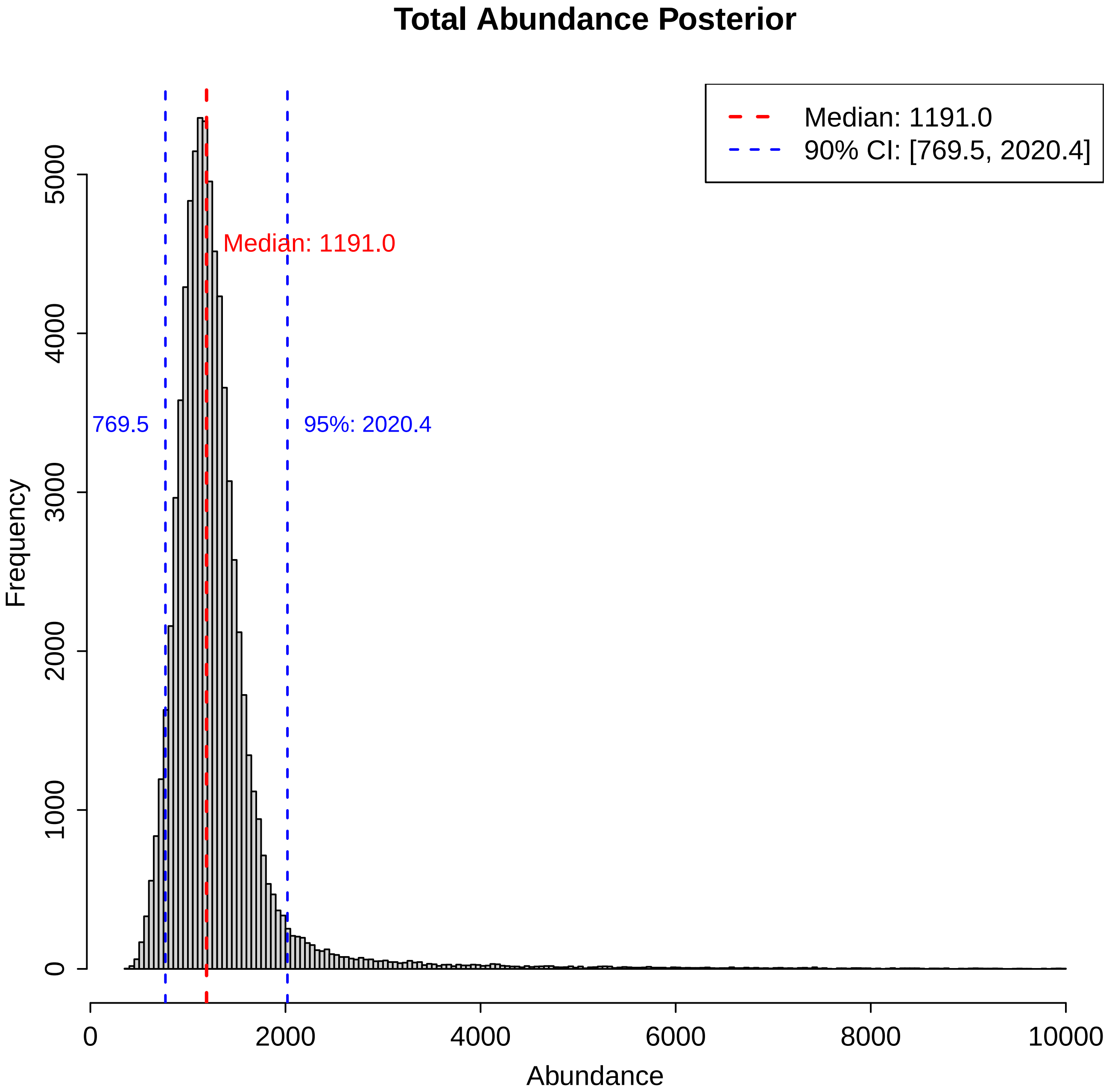
# SD Grid Group Abundance (z)





Transect Expected Abundance (lam)
lam
0
5
10
15
20
25
30
lam[1]
lam[5]
lam[9]
lam[13]
lam[17]
lam[21]
lam[25]
lam[29]


Posterior of Average Group Size (AGS)
Frequency
0
1000
2000
3000
1.6
1.8
2.0
2.2
2.4
2.6
2.8
3.0
Group Size



Total Abundance Posterior
Median: 1191.0
90% CI: [769.5, 2020.4]
Median: 1191.0
769.5
95%: 2020.4
Frequency
0
1000
2000
3000
4000
5000
0
2000
4000
6000
8000
10000
Abundance